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Article

Pricing event risk: evidence from concave implied volatility curves

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Abstract

We document that implied volatility (IV) curves of short-term equity options frequently become concave prior to earnings announcements day (EAD), typically reflecting a bimodal risk-neutral distribution for the underlying stock price. Firms with concave IV curves exhibit significantly higher absolute stock returns on EAD and higher realized volatility after the announcement, rendering concavity an ex-ante signal for event risk. Returns on delta-neutral straddles, delta-neutral strangles, and delta- and vega-neutral calendar straddles are negative and significantly lower in the presence of concave IV curves, showing that investors pay a substantial premium to hedge against the gamma risk arising from this event.

Keywords: earnings announcement; event risk; risk-neutral distribution; implied volatility.

1. Introduction

Earnings announcements are scheduled corporate events that disseminate substantial fundamental information to investors. A voluminous literature has examined several features, such as the behavior of stock returns (see, e.g., Ball and Brown 1968; Beaver 1968; Ball and Kothari 1991; Frazzini and Lamont 2007) and systematic risk (see, e.g., Patton and Verardo 2012; Savor and Wilson 2016) around these earnings announcements days (EADs).

We posit that these scheduled announcements are often viewed as referendums on firm value. On these occasions, investors anticipate that the underlying stock price will, more likely than not, exhibit a large movement in either direction upon the announcement. This anticipated stock price jump may induce bimodality in the *central part* of the ex-ante risk-neutral distribution (RND) and concavity in the implied volatility (IV) curve. Using data on very short-term options, we first empirically document that concavity in the IV curve is a pervasive feature prior to EADs and then study the pricing implications of this phenomenon.

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The possibility of stock price jumps can also translate into increased volatility around EADs. In fact, Dubinsky et al. (2019, DJKS henceforth) and Patell and Wolfson (1979, 1981) document an increase in IV in the runup to EADs and a sharp drop afterwards. However, bimodality is a fundamentally different concept of risk relative to a more dispersed distribution (volatility), or a negatively skewed distribution (a reflection of tail risk), or a more fat-tailed distribution (kurtosis).

Bimodality in the central part of the RND implies that, subject to a relatively small risk adjustment due to the very short option expiry, the prevailing stock price is expected to be around either of the two identified modes. This means that the stock price after the announcement will most likely be x percent above or y percent below the current price; each outcome may also be associated with a different volatility level. This feature is, therefore, also different from the common modeling assumption of a low-probability, randomly timed Poisson jump (see, e.g., Merton 1976; Ball and Torous 1985), which can lead to an IV smirk and a left-tailed RND, capturing tail risk and explaining the expensiveness of OTM puts (see, e.g., Bates 1996, 2000; Pan 2002; Yan 2011). We call this ex-ante bimodality as "event risk" for the underlying stock and argue that a concave IV curve provides an option-based signal for this type of risk.[1]

We show that during our sample period (2013–2020), a large fraction (38 percent) of IV curves extracted from short-expiry equity options become concave prior to EADs. This compares to just 4.8 percent of IV curves exhibiting concavity when option expiry does not span an EAD. The concave IV curves that we document are typically inverse U-shaped, S-shaped, or W-shaped. These shapes are in stark contrast with the convex volatility smiles and smirks (or skews) that are commonly observed for equity options, where out-of-the-money (OTM) puts trade at higher volatility relative to at-the-money (ATM) options. Interestingly, the feature of concavity mostly disappears right after the announcement, as the uncertainty surrounding this event is resolved, and the IV curves revert to their standard convex shape.

We document that concavity in the IV curve is typically associated with bimodality in the central part of the corresponding RND. Specifically, we find that 86 percent of the observations with concave IV curves exhibit a bimodal RND. While concavity does not axiomatically reflect a bimodal distribution, we show that a concave IV curve provides an option-based signal of impending event risk in the underlying stock.

Concavity appears in short- and not long-expiry options. This feature arises due to the relative effect between the anticipated jump and the diffusion component of the underlying stock price process. As expiry shrinks, the effect of the anticipated jump dominates the effect of the diffusion component; this renders the underlying RND bimodal and the IV curve concave. On the other hand, as the expiry increases, the diffusion component dominates, the RND reverts to unimodality and the IV curve to convexity.[2]

To rule out an alternative hypothesis that bimodal RNDs are driven by option traders erroneously understating the probability of no news, we examine the distribution of realized stock returns on EAD. We find that the distribution of EAD stock returns following a non-concave IV curve is unimodal (with mode around 0 percent)—similar to the distribution of daily stock returns on typical trading days. In contrast, following the formation of concave IV curves, the central part of the corresponding stock return distribution exhibits bimodality, with two distinct modes away from 0 percent. This finding confirms that bimodal RNDs arise primarily because large stock returns in either direction are highly likely to occur on EADs and investors price these anticipated outcomes in the option market.

[1] According to Liu, Longstaff, and Pan (2003, p. 231), event risk is defined as "the risk of a major event precipitating a sudden large shock to security prices and volatilities."

[2] Our analysis uses options with expiry between three and thirteen calendar days ahead. About 80 percent (20 percent) of our observations are constructed using weekly (regular) options. The sparsity of short-term equity options prior to our sample period may explain why this feature has not been previously documented in the literature.

Having documented these novel features of IV curves around EADs, we examine the informational content of concavity with respect to the pricing of event risk. We begin our analysis by providing further evidence that this feature is a valid ex-ante signal for event risk. First, we find that, on average, firms exhibiting concave IV curves have an absolute abnormal stock return of 5.88 percent on EAD, which is 1.64 percent higher than the corresponding absolute return for firms with non-concave IV curves. Second, we find that firms with concave IV curves exhibit an average realized stock return annualized volatility of 47.5 percent in the 10-day interval after the announcement, which is 10.38 percent higher than the corresponding realized volatility of firms with non-concave IV curves. These findings show that investors are able to identify earnings announcements that trigger larger than average stock price movements and volatility. Anticipating these effects, investors may trade accordingly in the option market, giving rise to concave IV curves, which in turn signal ex-ante the impending event risk.

The most obvious way investors could speculate on, or hedge against, large stock price swings on EADs, regardless of their direction, is by purchasing straddles. Delta-neutral ATM straddles have been commonly used to capture the price of volatility risk for the underlying stock returns (Coval and Shumway 2001). Therefore, we examine whether delta-neutral straddle returns on EADs differ across concave and non-concave IV curves. Interestingly, concave IV curves are followed by a negative and 4.65 percent lower average delta-neutral straddle return on EAD, as compared to non-concave IV curves. In fact, we find that only in the presence of concave IV curves do investors pay a significant premium to hedge against the uncertainty caused by the forthcoming announcement.

Delta-neutral straddles are exposed to both stochastic volatility (vega) and jump (gamma) risk. To identify which of these two sources of risk is priced around earnings announcements, we follow two complementary approaches. First, similar to Dew-Becker, Giglio, and Kelly (2021), we construct strangles that yield positive payoffs only when the underlying stock price exhibits a sufficiently large move. Hence, strangle returns can provide direct evidence on the price of gamma risk around EADs. Second, following Cremers, Halling, and Weinbaum (2015), we construct delta- and vega-neutral calendar straddles (which expose investors to gamma risk only) and delta- and gamma-neutral calendar straddles (which expose investors to vega risk only).

We find that, on average, concave IV curves are followed by a 9.05 percent lower strangle return and a 12.98 percent lower delta- and vega-neutral straddle return on EADs, as compared to non-concave IV curves. In fact, the average returns of these option strategies are negative only in the presence of concave IV curves. On the other hand, delta- and gamma-neutral straddles yield a positive premium across concave and non-concave IV curves. These results show that investors pay a substantial premium to hedge against the gamma risk that arises due to the earnings announcement only in the presence of concave IV curves. They also show that the informational content of the concave IV curves is related to gamma rather than vega risk.

We subsequently examine whether the negative straddle and strangle premia we attribute to concave IV curves could be alternatively explained by other risk measures related to short-term IV, which is known to be substantially elevated in the run up to EADs (see, e.g., Patell and Wolfson 1979, 1981; DJKS 2019). To this end, in our regression analysis, we control the level and term structure of IV as well as the spread between the realized and IV. We confirm that concavity contains a distinct and significant predictive ability with respect to the premia of the option strategies examined. In contrast, our portfolio analysis further shows that the IV-based measures are not significantly related to the magnitude of the straddle and strangle premia earned on EADs. Moreover, we find that the significance of concavity remains intact when we additionally control for measures capturing information contained in the tails of the RND, such as the risk-neutral skewness and kurtosis of Bakshi, Kapadia, and Madan (2003).

Even though the main objective of our paper is empirical, to better understand the drivers of concave IV curves, we introduce an option pricing model building on DJKS (2019) and Piazzesi (2000). DJKS model EAD jump size to be normally distributed, leading to a large increase in short-term ATM IV and a downward sloping term structure of IV prior to the announcement. In contrast, we allow the jump size to follow a mixture of normal distributions. Although seemingly an innocuous modification, our assumption is more consistent with the different conceptual underpinnings of price jump risk and volatility risk. More importantly, our modeling assumption can naturally generate a bimodal RND. In this respect, our model is closer in spirit to studies that have used mixtures of log-normal distributions to empirically fit the RNDs for various assets prior to geopolitical events or policy decisions (see, e.g., Leahy and Thomas 1996; Melick and Thomas 1997; Mirkov, Pozdeev, and Söderlind 2016; Hanke, Poulson, and Weissensteiner 2018).

In our analysis of relative model fits, we find that our model can more accurately fit the actual ATM IV and the downward sloping term structure of IV values, and hence it can better capture the information content of short-expiry options prior to EAD. However, the DJKS (2019) model underestimates the magnitude of the anticipated price jump volatility, and hence it yields downward biased estimates of both the ATM IV and the term structure of IV, especially in the presence of concave IV curves.

Our model and empirical methodology could be applied to any type of scheduled announcement or event as long as this is expected to trigger a substantial shift in the underlying asset price. To illustrate, we consider scheduled macroeconomic announcements. Prior to 2020, our sample period covers a relatively calm macroeconomic regime (low inflation, declining unemployment rate, and stable GDP growth). As a result, we find that, on average, the fraction of concave IV curves observed prior to these announcements is similar to the one observed on a typical trading day. Nevertheless, we identify several announcements, both in 2020 and before, which were indeed preceded by a large fraction of concave IV curves.

We further examine the behavior of the aggregate fraction of concave IV curves, excluding those that span EADs, and document a substantial time-series variation. The fraction of concave IV curves strongly aligns with the subsequent incidence of stock price jumps, indicating that concave IV curves represent ex-ante anticipation of stock price jumps beyond earnings announcements too. We also find that the fraction of concave IV curves is strongly positively correlated with macroeconomic and financial uncertainty indices of Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021), but not to the analyst forecast disagreement measure of Yu (2011), indicating that concavity in the IV curve primarily reflects a higher degree of uncertainty rather than differences in opinion.

Overall, our study shows that large stock price movements are systematically anticipated by investors prior to scheduled announcements and can be detected ex-ante because they dramatically affect the pricing of short-expiry options. In the case of concave IV curves, we show that large stock price movements are not just a possibility due to the announcement, but rather a very likely outcome. This feature often gives rise to a bimodal short-term RND for the underlying stock price (and return), which is in stark contrast with the established paradigm in asset pricing that relies on unimodal return distributions.

We contribute to various strands of the literature. Starting from the early studies of Patell and Wolfson (1979, 1981), there is a growing literature showing that option-based measures embed significant information prior to earnings announcements (see, e.g., Amin and Lee 1997; Ni, Pan, and Poteshman 2008; Xing, Zhang, and Zhao 2010; Billings and Jennings 2011; Jin, Livnat, and Zhang 2012; Barth and So 2014; Gao, Xing, and Zhang 2018; Lei, Wang, and Yan 2020; Chen, Gan, and Vasquez 2023). We add to this literature by showing that the curvature properties of the IV curve contain significant predictive ability over straddle and strangle premia earned on EADs.

Our study is also related to the literature that uses option prices to extract information regarding firm value following corporate announcements, such as proposed merger and acquisition transactions (see, e.g., Barraclough et al. 2013; Borochin 2014; van Tassel 2016). However, the timing of these announcements and the successful completion of the proposed transactions are typically uncertain. In contrast, there is no uncertainty about the timing of the scheduled earnings announcements in our sample, allowing us to model their impact via deterministically timed jumps.

Our setup is closely related to that of DJKS (2019), who also examine the impact of predictably timed EAD stock price jumps on option pricing. However, their focus is on the term structure of ATM IV prior to the announcement, whereas we examine the curvature properties of the IV curve for short-term equity options. Importantly, in their model, the EAD jump size is assumed to be normally distributed and its mean is a transformation of its volatility. As a result, the only effect of this anticipated price jump is a large increase in short-term ATM IV, leading to a downward sloping term structure prior to the announcement. The distribution of stock prices remains unimodal and the jump has no effect on the curvature of the IV curve across moneyness levels. Therefore, the model of DJKS cannot reproduce the novel but pervasive empirical features we document in our study, namely concavity in the IV curve and bimodality in the RND of the underlying stock price prior to the announcement.

We also build upon the literature showing that stock prices do jump upon the release of news in the form of scheduled earnings announcements (see, e.g., Lee and Mykland 2008; Lee 2012; Kapadia and Zekhnini 2019; Todorov and Zhang 2025). This occurs because upon the announcement, a discrete amount of information is released over a vanishingly small period of time. When this information signals a shift in firm profitability or risk, the announcement can substantially impact its valuation. Contributing to this literature, we show that investors systematically anticipate these large stock price moves and that concavity in the IV curve serves as a proxy to detect them ex-ante.

Our findings are also consistent with the conclusion of Kapadia and Zekhnini (2019) and Begin, Dorion, and Gauthier (2020) that firm-level jump risk is priced. In fact, we show that investors are averse to the gamma risk that arises prior to earnings announcements and pay a significant premium to hedge it when they anticipate large stock price moves.

Methodologically, our study is also related to the literature that examines the asset pricing effects of macroeconomic announcements conveying information regarding the state of the economy (see, e.g., Savor and Wilson 2013; Ai et al. 2022; Wachter and Zhu 2022) and political events that may lead to a shift in government policy (see, e.g., Pástor and Veronesi 2013; Kelly, Pástor, Veronesi 2016). We show that the curvature properties of the IV curve can be informative regarding the pricing of the corresponding event risk and our model is directly applicable to these scheduled events.

To this end, our results can be informative for the growing literature that examines the timing of the resolution of uncertainty around scheduled macroeconomic or earnings announcements (see, e.g., Lucca and Moench 2015; Hu et al. 2022; Gao, Hu, and Zhang 2023). We show that the anticipation of stock price jumps in either direction is an important component of the priced uncertainty prior to scheduled events. Therefore, the curvature of the IV curve, reflecting the properties of the RND, can provide additional information regarding the degree and the timing of the resolution of "impact uncertainty" in the two-risk model of Hu et al. (2022).

Our findings are also complementary to the literature on volatility information trading (see, e.g., Ni, Pan, and Poteshman 2008). Specifically, we highlight the informational content of short-expiry options with respect to anticipated stock price jumps, in either direction, prior to scheduled announcements. We show that concave IV curves are related to the premium investors pay to hedge against the gamma (rather than vega) risk that arises from these announcements, buying, for example, short-expiry ATM straddles or strangles. Since jump risk cannot be perfectly hedged, market makers would require a premium to be counterparties in these trades (see, e.g., Figlewski 1989; Green and Figlewski 1999). Consistent with the demand-based option pricing framework of Gârleanu, Pedersen, and Poteshman (2009), this trading activity can lead to higher implied volatilities for the relevant range of strikes, giving rise to a concave IV curve for short-expiry options prior to scheduled announcements.

The remainder of the article is organized as follows. Section 2 describes the data and the methodology for calculating IV curves. We discuss the features of concave curves and their relation to bimodality in return distribution in Section 3. Section 4 presents our model and its comparison with that of DJKS (2019). The main empirical results are in Section 5 where we show that investors pay substantial premia to hedge gamma risks only in the presence of concave IV curves. Section 6 discusses the behavior of concave IV curves beyond earnings announcements. We conclude in Section 7.

2. Data and methodology

2.1 Option data and IV curves

We construct IV curves using option data from OptionMetrics during the period 2013 to 2020. For each calendar year, we select 100 firms with the highest option trading volume, requiring the underlying to be common stock (share codes 10 or 11) with share price higher than \$5. This yields a total sample of 194 firms during the entire period. The choice of the sample period and the cross-section of firms are dictated by the availability and liquidity of short-term options data that we require for our analysis.[3] Weekly equity options have been actively traded for a range of strikes only in the last decade. Hence, OptionMetrics provides very sparse data for short expiries with a sufficient number of strikes prior to 2013.[4]

Our primary focus is on option-implied information related to earnings announcements, so we utilize short-term options whose expiration spans the EAD. In particular, we keep options with expiry between three and thirteen calendar days ahead. Although our sample predominantly consists of weekly options, we also keep regular options as long as they expire between three and thirteen calendar days ahead. The short expiry options we employ are very liquid on the trading day prior to EAD. Specifically, the total trading volume across all strikes is on average (median) 36,555 (18,096) contracts. To put this figure into perspective, we compare it with the corresponding total trading volume on the same day of the next option with at least 30 days to expiry. We find that the total trading volume of the short expiry we use is on average (median) 4.0 (2.5) times higher than the corresponding volume of the expiry with at least 30 days. Therefore, the short-expiry options we use are more liquid prior to the EAD relative to the options with standard expiration often used in the literature.

To ensure that the information embedded in the IV curves is meaningful, we apply a number of standard filters to the option data. Specifically, we discard options with zero open interest, zero trading volume, zero bid price, bid higher than ask price, mid-quote price less than \$0.125, non-standard settlement, or missing IV. To make sure that our findings are not driven by particularly illiquid strikes, we also discard options when the bid–ask spread is higher than 20 percent of the mid-quote price.

We obtain information on the timing of quarterly EADs from I/B/E/S. Following common practice in the literature (see Barth and So 2014; Michaely, Rubin, and Vedrashko 2014),

[3] Our sample firms represent, on average, 44 percent of the total market value across all US common stocks. This is not a surprise because we select the firms with the highest option trading volume per year, which are usually the largest capitalization firms. An added advantage of our (relatively small) cross-section is that our results are not driven by small-capitalization firms.

[4] For example, after imposing liquidity filters and minimum number of strikes (to be described below) we have two firm-quarter observations in 2006, and still only fifty-nine observations in 2012.

if the announcement is made after the market close, the next trading day is defined as the EAD.

To construct the IV curve, we utilize the annualized IVs of ATM and OTM options provided by OptionMetrics. To avoid an artificial jump at the ATM region, which could arise from ATM puts potentially trading at higher IV relative to ATM calls, we follow the blending approach of Figlewski (2010). Specifically, we blend the IVs of puts (IV_P) and calls (IV_C) whose strike price K lies within ± 2 percent of the underlying spot price, S, into a single point as follows:

$$IV(K) = wIV_P(K) + (1-w)IV_C(K), \tag{1}$$

where $w = (K_{high} - K)/(K_{high} - K_{low})$ and K_{high} (K_{low}) is the highest (lowest) strike in this ± 2 percent range. To ensure a good coverage of the moneyness range, after the blending we require at least six options for a given expiry, with at least two puts and two calls.

Equipped with these IV points, we fit a quintic spline using the function spaps in MATLAB.[5] This yields the smoothest IV curve in the moneyness space K/S, subject to a tolerance level for the sum of squared errors between the actual and the fitted IVs. In the spirit of Bliss and Panigirtzoglou (2002, 2004), the quintic spline minimizes the following objective function:

$$\rho \sum_{i=1}^{N} [IV(K_i) - \widehat{IV}(K_i;\theta)]^2 + \int_{-\infty}^{\infty} S^{(3)}(x;\theta)^2 dx, \tag{2}$$

where $IV(K_i)$ is the actual IV for strike K_i, $\widehat{IV}(K_i;\theta)$ is the corresponding fitted IV, which is a function of the parameter set θ that defines the quintic spline $S(\theta)$, and ρ is a smoothing parameter that is optimally selected to ensure that the sum of squared IV errors does not exceed a given tolerance level.[6]

To compute the RND corresponding to the fitted IV curve, we use the standard result of Breeden and Litzenberger (1978). The density function is given by $f(K) = e^{rT}\partial^2 C/\partial K^2$, where r is the interest rate and C is the call option price as a function of the strike price K. The fitted IV curve contains 1,001 points. These IVs are converted to call option prices using the Black–Scholes formula. In the absence of a continuum of strikes, we compute the second partial derivative in the above formula using finite differences and derive the RND for the range of the available moneyness levels.

Having imposed a number of strict filters on the option data, we seek to fit well the actual IV points, and hence we opt for a low tolerance level. This tolerance level corresponds to a 0.01 percent root mean squared error between the actual and the fitted IVs. However, to ensure that the fitted IV curve is not too erratic and does not correspond to an ill-behaved RND, we impose further conditions. We require that no interpolated IV point is negative and that the corresponding RND does not exhibit a negative density point or more than two modes. If any of these conditions is violated, we increase the upper bound of the root mean squared error in steps of 0.005 percent until the conditions are met. Our final sample consists of 2,229 IV curves on the trading day prior to EAD for the firms in our sample. About 80 percent (20 percent) of these observations are constructed using weekly (regular) options.

5 A quintic spline ensures that the third derivative of the IV curve (and hence the option price function) is continuous, yielding a well-behaved RND (see Figlewski 2010).

6 Parameter ρ controls the tradeoff between the goodness-of-fit and the smoothness of the spline function; the latter is captured by its integrated squared third derivative. Setting a low tolerance level ensures that the spline fits well the actual IV points at the expense of smoothness. To the contrary, setting a high tolerance level yields a rather smooth spline that may not fit well the actual IV points.

2.2 Definition of concave IV curve

We introduce a definition of concavity based on the first and second derivatives of the fitted IV curve with respect to moneyness.[7] Specifically, we define an IV curve to be concave when the following three conditions hold. First, the second derivative of the fitted IV curve is negative for a continuous moneyness (K/S) range of at least 0.03 points, that is, for a continuous range of strikes that amount to at least 3 percent of the underlying spot price. Second, we require that the fitted IV curve exhibits a stationary point within the moneyness range where it exhibits concavity. Third, this stationary point is located between the second lowest ($K_{\min+1}$) and second highest ($K_{\max-1}$) strikes of the actual IV points used to fit the smooth IV curve.

These conditions address the potential concern that the documented concavity may be an artifact of outliers or the employed smoothing spline. In particular, they ensure that our definition does not simply capture very local inflection points or marginally concave parts of the IV curve. They also ensure that concavity does not arise from the lowest or highest actual strikes, which typically correspond to deep OTM options.

This definition is sufficiently general to capture various shapes of concavity, such as the inverse U-shape, W-shape, and S-shape IV curves illustrated in figure 1. Using this definition, we define the indicator variable `CONCAVE`, which takes the value one when the IV curve is concave and zero otherwise.

2.3 Other variables and data sources

In addition to `CONCAVE`, we use a number of other variables in the subsequent empirical analysis. The definition of these variables is provided in Appendix A. For each firm, at the daily frequency, we compute its market beta (`BETA`), the natural logarithm of market capitalization (`Ln(SIZE)`) and stock price (`Ln(PRICE)`), the cumulative stock return of five days (`RUNUP`), momentum return (`MOM`), and stock turnover ratio (`STOCKTR`). We obtain stock prices, trading volumes, and number of outstanding shares from CRSP. We compute the book-to-market ratio (`B/M`) using quarterly data from COMPUSTAT. We also use the number of analysts providing earnings forecasts (`NUMEST`), the standard deviation of these forecasts (`DISP`), and the differential stock beta around EADs (`ANNBETA`) as in Barth and So (2014). Analysts forecast data are obtained from I/B/E/S.

We also use a number of option-based variables, such as the ATM IV (`ATMIV`) and the difference (`RVIV`) between the realized volatility and `ATMIV` of Goyal and Saretto (2009). Since our focus is on short-expiry options, we construct `ATMIV` and `RVIV` utilizing the 10-day volatility surfaces that have been recently introduced by OptionMetrics.[8] In addition, we compute the risk-neutral skewness (`RNS`) and risk-neutral kurtosis (`RNK`), following the approach of Bakshi, Kapadia, and Madan (2003). We also use the option-to-stock trading volume ratio (`O/S`) of Roll, Schwartz, and Subrahmanyam (2010). Finally, we compute the term structure estimate of ATM IV (`TSIV`) proposed by DJKS (2019).

2.4 Summary statistics

Table 1 presents the summary statistics for the variables used in our analysis. Their values are computed on the day prior to EAD and they are winsorized at the 1 percent and 99 percent levels. We find that 38.4 percent of the IV curves extracted prior to the EAD exhibit concavity. These IV curves are calculated from short-expiry options, with an average `EXPIRY` of 6.46 calendar days and a large number of strikes (average `STRIKES` = 17.88). The latter feature is consistent with the fact that our sample consists of very large firms, with an average (median) market capitalization of \$58 (\$68) billion. As a result, these firms

7 First and second derivatives of the fitted IV curve are computed using finite differences.

8 There is no material change in any of the results in Section 5 if, instead of `ATMIV`, we use the actual ATM implied volatility computed from the corresponding short-expiry option price or the risk-neutral variance of Bakshi, Kapadia, and Madan (2003), computed again from the actual prices of the corresponding short-expiry options. Both of these variables are very highly correlated with `ATMIV`.

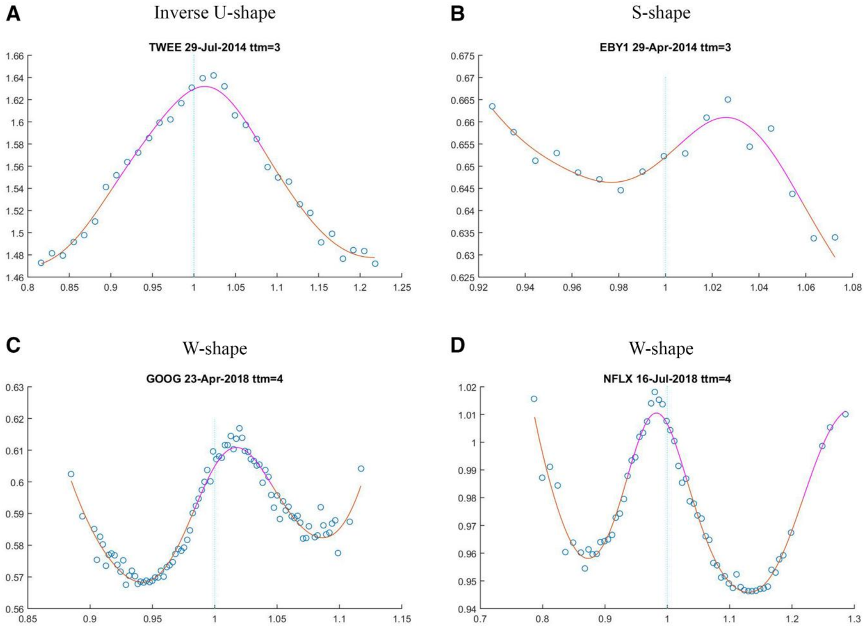

Figure 1. Types of concave IV curves. This figure shows different types of concave IV curves computed on the day prior to the EAD. Panel A presents an example of an inverse U-shape IV curve for Twitter, computed from options with 3 days to expiry on July 29, 2014. Panel B presents an example of an S-shape IV curve for Ebay, computed from options with 3 days to expiry on April 29, 2014. Panels C and D present examples of W-shape IV curves for Google and Netflix, respectively, computed from options with 4 days to expiry on April 23 and July 16, 2018, respectively. Circles indicate implied volatilities corresponding to actual traded strikes, whereas the curve is fitted using a quintic spline.

trade at a much higher price (average = \$77.48), they exhibit low B/M (average = 0.35), and are followed by a very large number of analysts (average NUMEST = 23.95), as compared to the corresponding values typically encountered in studies that use the entire CRSP universe.

Regarding option-based variables, the median RNS (RNK) is − 0.25 (3.46). In line with the arguments of Patell and Wolfson (1979, 1981), ATMIV is substantially higher prior to EADs, with an average value of 45.04 percent. As a consequence, RVIV takes very large negative values, with an average of − 16.62 percent. Moreover, TSIV is almost always positive, with an average value of 6.80 percent. This confirms the findings of DJKS (2019) that the term structure of ATM IV is downward sloping prior to EADs. Lastly, we also find substantial stock trading activity prior to EAD, with an average daily STOCKTR of 2.4 percent, and an even higher trading activity in the option market, with an average O/S of 28.43 percent.

Table 2 reports the pairwise correlations between these variables. Our main focus is on the correlation properties of the newly proposed variable CONCAVE. In particular, we find that CONCAVE is positively correlated with ATMIV, RNS, and TSIV, but negatively correlated with RNK and RVIV. Hence, concave IV curves are associated with higher levels of ATM IV and a steeper downward-sloping IV term structure prior to EAD. Moreover, CONCAVE exhibits a positive correlation with STOCKTR, O/S, and NUMEST, indicating that concave IV curves appear more often when there is substantial coverage by financial analysts as well as high trading activity from investors prior to the announcement.

We note that the reported correlations for CONCAVE are not particularly high (much less than 0.40 in absolute value), alleviating the potential concern that CONCAVE may simply

Table 1. Summary statistics.

This table presents summary statistics for selected variables. CONCAVE is an indicator variable that takes the value one when the IV curve is concave on the day prior to the EAD and zero otherwise. ABSEADABRET is the absolute abnormal stock return on EAD, measured with respect to the four-factor FFC model. POSTEADVOL is the 10-day post-EAD annualized realized stock return volatility. STRADDLE denotes the return of the delta-neutral ATM straddle strategy on EAD. IMPMOVE denotes the ratio of the sum of the ATM put and call prices divided by the underlying stock price. STRANGLE denotes the return of the delta-neutral strangle strategy on EAD. JUMPSTRADDLE denotes the return of the delta- and vega-neutral ATM straddle strategy on EAD. VOLSTRADDLE denotes the return of the delta- and gamma-neutral ATM straddle strategy on EAD. The definition of the rest of the variables is provided in Appendix A. These summary statistics are computed for a sample of quarterly earnings announcements during the period 2013–2020.

Variable	Mean	St. Dev.	25th pctl	Median	75th pctl	Obs.
CONCAVE	0.384	0.49	0	0	1	2,229
EXPIRY	6.46	2.60	4	8	9	2,229
STRIKES	17.88	12.85	9	14	22	2,229
BETA	1.09	0.31	0.89	1.09	1.27	2,188
Ln(SIZE)	10.96	1.33	10.07	11.13	12.00	2,220
B/M	0.35	0.33	0.12	0.25	0.46	2,085
RUNUP	0.68	4.36	−1.63	0.64	2.85	2,229
MOM	18.73	46.80	−6.89	12.00	32.29	2,188
Ln(PRICE)	4.35	0.92	3.75	4.22	4.83	2,229
ATMIV	45.04	22.41	29.24	37.74	55.33	2,177
RNS	−0.25	0.28	−0.42	−0.25	−0.09	2,229
RNK	3.63	0.65	3.24	3.46	3.81	2,229
RVIV	−16.62	15.03	23.61	14.72	7.39	2,177
TSIV	6.80	3.79	4.02	5.73	8.65	2,217
NUMEST	23.95	7.66	18	23	29	2,219
DISP	14.70	30.21	2.56	4.84	12.00	2,209
ANNBETA	0.10	0.78	−0.29	0.07	0.49	2,112
STOCKTR	2.40	3.20	0.66	1.16	2.74	2,229
O/S	28.43	32.75	6.13	16.55	37.26	2,229
ABSEADABRET	4.86	4.72	1.60	3.41	6.36	2,188
POSTEADVOL	41.09	26.71	22.48	33.39	51.65	2,227
STRADDLE	−0.86	49.00	−33.65	−15.43	18.37	2,181
IMPMOVE	6.53	3.38	4.08	5.55	8.12	2,181
STRANGLE	−2.32	80.22	−55.71	−28.78	25.54	1,909
JUMPSTRADDLE	−2.82	137.08	−89.79	−39.74	47.69	1,888
VOLSTRADDLE	1.17	4.35	−1.41	0.81	3.38	1,895

mimic another firm characteristic. In contrast, Table 2 shows very high pairwise correlations between ATMIV, RVIV, TSIV, and Ln(SIZE) prior to EADs (e.g., the correlation between ATMIV and TSIV is 0.92).

Table 3 compares the average values of these variables between the observations of concave and non-concave IV curves on the day before EAD. We find that concave IV curves are extracted from sets of options with a somewhat shorter average expiry and a higher average number of available strikes. We also find that concave IV curves are associated with firms that, on average, are followed by more analysts, they are relatively smaller, and they have lower B/M.

In addition, we observe that concave IV curves are associated with significantly higher average values of BETA, STOCKTR, and O/S as well as higher average stock prices and returns (Ln(PRICE), RUNUP, MOM) prior to the EAD. Consistent with the pairwise correlations presented in Table 2, we also report that concave IV curves are accompanied, on average, by significantly higher ATMIV, RNS, and TSIV values and significantly lower RNK and RVIV values relative to non-concave IV curves.

Table 2. Pairwise correlations of firm characteristics.

This table presents pairwise correlation coefficients among selected variables. CONCAVE is an indicator variable that takes the value one when the IV curve is concave on the day prior to the EAD and zero otherwise. The definition of the rest of the variables is provided in Appendix A. These correlations are based on the values of the variables measured on the day prior to the EAD and they are computed for a sample of quarterly earnings announcements during the period 2013–2020.

	CONCAVE	BETA	Ln(SIZE)	B/M	RUNUP	MOM	Ln(PRICE)	ATMIV	RNS	RNK	RVIV	TSIV	NUMEST	DISP	ANNBETA	STOCKTR
BETA	0.09															
Ln(SIZE)	−0.18	−0.26														
B/M	−0.11	0.18	−0.13													
RUNUP	0.02	0.04	−0.02	0.04												
MOM	0.11	0.11	−0.03	−0.24	0.06											
Ln(PRICE)	0.10	−0.10	0.42	−0.35	0.07	0.22										
ATMIV	0.31	0.29	−0.63	−0.03	0.02	0.14	−0.19									
RNS	0.33	0.04	−0.17	0.02	0.08	0.10	0.02	0.13								
RNK	−0.31	−0.07	0.30	0.04	0.06	−0.02	0.15	−0.13	−0.27							
RVIV	−0.30	−0.12	0.41	0.09	0.04	−0.11	0.09	−0.54	−0.18	0.22						
TSIV	0.36	0.24	−0.59	−0.14	0.02	0.18	−0.13	0.92	0.13	−0.24	−0.61					
NUMEST	0.19	0.04	0.23	−0.25	0.02	0.06	0.28	0.05	0.05	−0.10	−0.15	0.17				
DISP	0.01	0.15	−0.23	0.10	0.02	0.02	−0.03	0.32	0.01	0.07	−0.14	0.24	−0.12			
ANNBETA	0.05	0.12	−0.09	0.00	0.02	0.09	0.02	0.13	0.01	−0.02	−0.10	0.14	−0.04	0.05		
STOCKTR	0.25	0.27	−0.62	−0.01	0.10	0.24	−0.08	0.74	0.11	−0.13	−0.41	0.73	−0.00	0.26	0.14	
O/S	0.24	−0.07	0.05	−0.13	0.07	0.09	0.50	0.03	0.12	0.06	−0.12	0.08	0.14	−0.03	0.05	0.08

Table 3. Characteristics of firms with concave versus non-concave IV curves.

This table presents the average values of selected variables for firms when they exhibit a concave IV curve on the day prior to the EAD (CONCAVE = 1) versus the corresponding average values when they do not exhibit a concave IV curve (CONCAVE = 0). ABSEADABRET is the absolute abnormal stock return on EAD, measured with respect to the four-factor FFC model. POSTEADVOL is the 10-day post-EAD annualized realized stock return volatility. STRADDLE denotes the return of the delta-neutral ATM straddle strategy on EAD. IMPMOVE denotes the ratio of the sum of the ATM put and call prices divided by the underlying stock price. STRANGLE denotes the return of the delta-neutral strangle strategy on EAD. JUMPSTRADDLE denotes the return of the delta- and vega-neutral ATM straddle strategy on EAD. VOLSTRADDLE denotes the return of the delta- and gamma-neutral ATM straddle strategy on EAD. The definition of the rest of the variables is provided in Appendix A. The last column contains the difference in the average values with the corresponding *t*-statistic (under the null hypothesis of equal means) in parenthesis. The sample consists of quarterly earnings announcements during the period 2013–2020.

Variable	CONCAVE = 1	CONCAVE = 0	Difference
EXPIRY	6.14	6.65	−0.52 (4.54)
STRIKES	22.15	15.22	6.93 (11.88)
BETA	1.12	1.07	0.05 (3.51)
Ln(SIZE)	10.67	11.14	−0.47 (−7.90)
B/M	0.31	0.38	−0.08 (−5.33)
RUNUP	0.94	0.52	0.42 (2.19)
MOM	24.95	14.93	10.02 (4.55)
Ln(PRICE)	4.45	4.29	0.16 (3.68)
ATMIV	53.41	39.85	13.56 (14.07)
RNS	−0.14	−0.32	0.19 (17.72)
RNK	3.37	3.80	−0.43 (−17.21)
RVIV	−22.24	−13.13	−9.11 (−13.61)
TSIV	8.42	5.79	2.63 (16.42)
NUMEST	25.57	22.94	2.62 (7.68)
DISP	15.31	14.31	1.00 (0.78)
ANNBETA	0.14	0.08	0.06 (1.54)
STOCKTR	3.24	1.88	1.36 (9.40)
O/S	37.84	22.58	15.27 (10.10)
ABSEADABRET	5.88	4.24	1.64 (7.71)
POSTEADVOL	47.49	37.11	10.38 (8.82)
STRADDLE	−3.74	0.91	−4.65 (−2.16)
IMPMOVE	7.89	5.69	2.20 (15.24)
STRANGLE	−7.94	1.12	−9.05 (−2.45)
JUMPSTRADDLE	−10.69	2.29	−12.98 (−2.12)
VOLSTRADDLE	1.32	1.08	0.25 (1.19)

3. Features of concave IV curves

IV curves for equity options typically exhibit a smile or a smirk (see, e.g., Rubinstein 1994; Toft and Prucyk 1997; and the review of the early literature in Jackwerth 2004), which corresponds to a convex IV curve where OTM puts trade at higher IV than ATM options. This pattern corresponds to an important deviation from the Black and Scholes (1973) model, where IV should be constant across moneyness levels. In sharp contrast to the commonly documented convex IV curves for equity options, as shown in the summary statistics, we often observe concave IV curves prior to EADs. This section illustrates the main features of the concave IV curves observed in the data.[9]

[9] The unusual shape for the IV curve has also attracted the attention of practitioners. For example, Baker, Gillberg, and Thomas (2018) show that the IV curve computed from options on GBPUSD futures became non-convex prior to the 2016 Brexit referendum. They also quote anecdotal evidence from Vola Dynamics LLC that this phenomenon can be observed for single-stock options prior to earnings announcements. This is consistent with our systematic evidence that concavity in the IV curve of short-expiry equity options is a pervasive feature prior to firms' EADs.

Figure 1 provides examples of the three main types of concavity we encounter in our sample. In this figure, circles indicate implied volatilities corresponding to actual traded strikes, whereas the curve is fitted using a quintic spline. Panel A shows an inverse U-shape IV curve for Twitter, computed from options with three days to expiry on July 29, 2014. Here, the IV of OTM calls and puts is substantially lower than the IV of ATM options. Panel B illustrates an S-shape curve for eBay, computed from options with three days to expiry on April 29, 2014. This curve exhibits two stationary points. In this particular example, the concave part of the curve is located in the OTM calls region, whereas the OTM puts region exhibits a typical convex shape. An interpretation of this shape is that concavity arises in a specific moneyness range, where options are trading at higher volatility relative to neighboring strikes. Panels C and D provide examples of an even more intriguing type of concavity, for Google and Netflix, respectively, computed from options with four days to expiry on April 23 and July 16, 2018, respectively. This W-shaped IV curve exhibits three stationary points, with a U-shape curve followed by an inverse U-shape curve, which is in turn followed by another U-shape curve. Here, concavity arises in specific ranges of moneyness, with near-the-money options trading at volatility levels as high as or even higher than deep OTM options.

The above concavity shapes appear systematically in short-expiry equity options just before EADs. We find that these shapes typically disappear right after the announcement, with the IV curve reverting to a standard convex shape. Figure 2 illustrates this pattern using, as an example, the Apple earnings announcement that occurred right after the market close on October 28, 2013. Whereas the IV curve extracted just before the announcement from options with four days to expiry exhibits a clear W-shape, it reverts to a smile on the following day using options with the same expiry date.

Figure 3 further illustrates that the IV curves often become concave in the runup to the EAD, but they subsequently revert to their standard convex shape. Specifically, figure 3 reports the fraction of concave IV curves for the firms in our sample on trading days around the EAD d. We observe that the fraction of concave IV curves gradually increases from 20 percent on day $d-5$ to 26.9 percent on day $d-2$, reaching a peak of 38.4 percent on the trading day prior to EAD. Right after the announcement, there is a sharp drop in the fraction of IV curves exhibiting concavity to only 8.7 percent on day d. This fraction subsequently drops further and hovers around 5 percent from day $d+1$ onwards.

3.1 Is concavity spurious?

To emphasize how uncommon it is to find a large fraction of concave IV curves using options whose expiration does not span an EAD, we perform the following analysis. For the firms in our sample, we impose the same data filters and follow the same steps of the methodology described in Section 2.1 to compute CONCAVE on all trading days during the period 2013–2020. We extract 90,464 firm-day IV curves from very short-term options whose expiry does not span an EAD. We find that only 4.8 percent of these observations exhibit a concave IV curve. This finding alleviates the potential concern that the large fraction of concave IV curves we identify in the run up to the EAD may be an artifact of our methodology or the use of very short-expiry options. Nevertheless, our definition of concavity in Section 2.2 does require a few empirical choices. In this subsection, we address potential concerns with our approach.

3.1.1 Blending of IVs

A potential concern is that concavity might be an artifact of blending OTM puts and calls around the ATM region, following the approach of Figlewski (2010). To alleviate this potential concern, we alternatively compute IV curves of puts and calls (using OTM, ATM, and ITM options) separately. We find that concavity is present in the IV curves computed from puts and calls separately. Therefore, the concavity feature does not arise due to our blending approach.

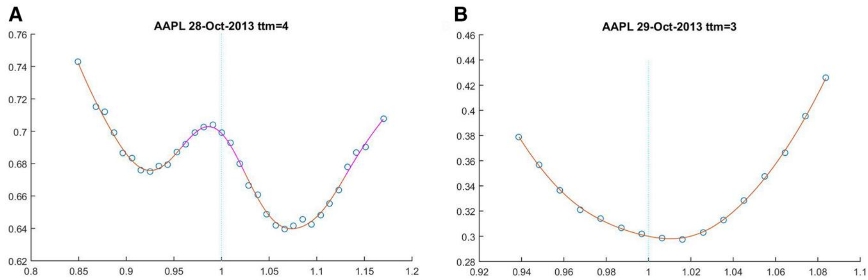

Figure 2. Concave IV curves around EAD. This figure illustrates how a concave IV curve prior to the EAD becomes convex after the announcement. The left panel presents a concave IV curve for Apple, computed from options with 4 days to expiry on October 28, 2013, that is, prior to its quarterly earnings announcement. The right panel presents a convex IV curve for the same firm, computed from options with 3 days to expiry on October 29, 2013, that is, right after the announcement.

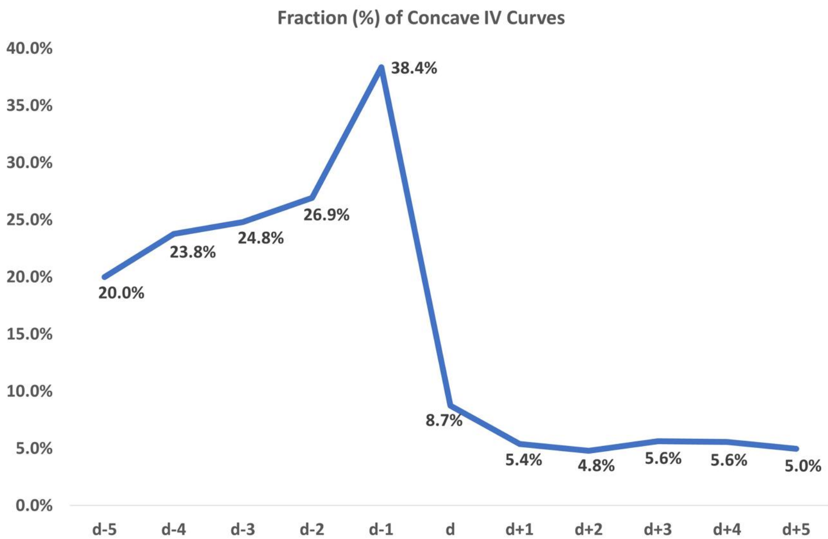

Figure 3. Fraction of concave IV curves around EAD. This figure shows the fraction of firms exhibiting a concave IV curve on each trading day from $d-5$ to $d+5$, where d is the quarterly EAD. The definition of a concave IV curve is provided in Section 2.2. IV curves are computed for the 100 firms with the highest option trading activity per year during the period 2013–2020.

3.1.2 American options

Since we use American-style equity options, early exercise premium could affect the definition of concavity. However, our dataset consists of very short-expiry OTM and ATM options. As shown in the prior literature (see Bakshi, Kapadia, and Madan 2003; and the recent evidence in Bakshi, Cao, and Zhong 2021), the combination of these two characteristics ensures that the early exercise premium, and hence its impact on IVs, would be very small, if any. Nevertheless, we additionally filter out firms that pay dividends during the life of the options. Under standard assumptions, calls on non-dividend paying stocks should not be exercised early, and hence there should be no premium. Moreover, the risk-free rate during our sample period is very low, so the early exercise premium for puts should also be negligible. As expected, only 3.5 percent of the firms in our sample pay dividends during the life of these options. Excluding these firms, we still find that 38.9 percent of the IV curves exhibit concavity on the day prior to EAD.

3.1.3 Borrow fees

Muravyev, Pearson, and Pollet (2022) argue that in the presence of high stock borrow fees, IVs provided by OptionMetrics may be misleading because they are computed as if this fee is zero. This is unlikely to be an issue in our study because our sample consists of very large capitalization firms, which typically have low stock borrow fees. Nevertheless, to address this concern, we additionally filter out firms with high stock borrow fees. Specifically, we utilise the categorical variable Daily Cost of Borrowing Score (DCBS) provided by Markit. This variable takes values from 1 to 10, where 1 (10) indicates that a stock has the lowest (highest) borrow fee. Blocher and Whaley (2015) show that stocks with DCBS of 1 have a mean (median) fee of just 36 (27) bps per annum. Equipped with this variable, we drop firms with DCBS greater than 1 on the day prior to EAD. Hence, we are only left with firms exhibiting a negligible borrow fee. As expected, only 5 percent of the firms in our sample have a DCBS greater than 1. Excluding this small fraction of firms, we still find that 37.7 percent of the IV curves exhibit concavity the day prior to EAD.

3.1.4 Bid–ask spreads

How reliable is the concavity measure given the high transaction costs in option markets (Christoffersen et al. 2018)? First, recall that our sample consists of 100 firms with the highest option trading volume per calendar year. Therefore, our sample of very liquid options is by design the least likely to be affected by microstructure biases. Second, we impose a number of strict liquidity filters to our option sample. For example, we discard strikes where the quoted bid–ask spread is higher than 20 percent of the mid-price. Third, as discussed in Section 2.1, the short-expiry options in our sample are very liquid on the trading day prior to the earnings announcement.

Finally, note that the feature of concavity most commonly appears near the ATM region. This is also the moneyness region where option bid–ask spreads tend to be lower. In fact, for the subsample of concave IV curves, we find that the average (median) bid–ask spread is only 5.99 percent (5.13 percent) across options in the moneyness range 0.98–1.02, and 6.81 percent (5.83 percent) across options in the moneyness range 0.95–1.05. Therefore, we conclude that the bid–ask spreads for the short-expiry options used in our study are quite tight and similar in magnitude across concave and non-concave IV curves.

3.1.5 Non-synchronous trading

Another potential concern could be that IVs are computed from option prices that may be non-synchronously recorded across different moneyness levels. In that case, concavity might appear spuriously because of combining actively traded prices from the ATM region and potentially stale prices from the OTM region. We reiterate that the options used in our study are actively traded. For instance, to compute IV curves, we use only those strikes with a non-zero trading volume on the corresponding day. Nevertheless, to rule out this concern, we obtain high-frequency option prices from the CBOE for a small number of firms and compute their IV curves at different times of the day. For example, when we calculate the IV curve for Apple on October 28, 2013, at different times of the day (10 am, 11 am, 12 noon, 1 pm, 2 pm, and 3 pm), we find that it is indistinguishable from the one reported in figure 2. Although not a large-scale exercise, this evidence provides further reassurance that the documented concavity in the IV curve does not spuriously arise from the use of end-of-day option prices from OptionMetrics.

3.2 Bimodality in RND

The main variable of interest in our analysis (CONCAVE) is defined with respect to the properties of the IV curve. To an extent, the shape of the IV curve reflects the properties of the RND for the underlying stock price. For example, a symmetric volatility smile corresponds to a leptokurtic RND, whereas a volatility smirk (or skew) is typically associated with a negatively skewed RND (see the related discussion in Jackwerth 2004; Hull 2009, chapter 18).

Figure 4 presents an example of a concave IV curve reflecting a bimodal RND for the underlying stock price. This is a rather unusual feature. RND bimodality implies that at option expiration, the underlying stock will most likely (under the risk-neutral measure) trade around either of the two identified price modes. The right panel of figure 4 illustrates the RND for Amazon, extracted from options with eight days to expiry on April 26, 2018, that is, just before the earnings announcement that took place right after the market close. The closing stock price was $1,517.96 on that day. The 8-day RND reveals two price modes at expiry; one at $1,444.80 (i.e., 4.8 percent lower) and the other one at $1,602.00 (i.e., 5.5 percent higher). Following the announcement, Amazon's stock price had a positive return of 3.6 percent on April 27 and closed at $1,580.95 (i.e., 4.15 percent higher) on May 4, at option expiry.

An interpretation of RND bimodality prior to an EAD, as illustrated in figure 4, is that a discrete price movement or jump is anticipated due to the forthcoming announcement.[10] In fact, Hull (2009, p. 398) describes a concave inverse U-shape IV curve as a "frown" and argues that it reflects a bimodal RND for the underlying asset price, which in turn arises "when a single large jump is anticipated." Therefore, we argue that a bimodal RND and a concave IV curve provide option-based signals of impending event risk in the underlying stock. Our analysis reveals that earnings announcements frequently give rise to event risk, which is priced in the option market and therefore can be detected ex-ante.

Although concavity in the IV curve does not axiomatically reflect RND bimodality (see Glasserman and Pirjol 2023 for a formal treatment), we find that 86 percent of the observations with CONCAVE = 1 prior to EAD exhibit a bimodal RND.[11] To ensure that this bimodality feature captures distinct modes rather than local wiggles in the RND, we further require that the identified modes are located at least 5 percent apart in terms of moneyness. Using this stricter definition of bimodality, we still find that 80 percent of the observations with CONCAVE = 1 exhibit a bimodal RND. To even further ensure that the documented bimodality feature is not driven by the tails of the RND, we additionally require that neither of the identified modes is located outside the [0.85 1.15] moneyness range. Despite imposing this additional requirement, we still find that 79 percent of the observations with CONCAVE = 1 exhibit a bimodal RND.

RND bimodality is an important feature that distinguishes our study from DJKS (2019). DJKS's model allows for predictably timed price jumps on EADs. However, by assuming a normally distributed EAD jump size, their implied RND remains unimodal, and hence their model cannot reproduce the concave IV curves observed in the data.

We emphasize that concave IV curves appear predominantly in short-term expiration options. Figure 5 illustrates an example of fitted IV curves for Amazon across different expirations (8, 22, 36, and 50 days) on April 26, 2018. The figure shows that while the IV curve for the 8-day expiry clearly exhibits a W-shape type of concavity, this feature is much less obvious for the 22-day expiry and disappears for longer expirations.

10 We also contemplated the idea that a bimodal RND prior to EAD might reflect a bimodal distribution of analysts' EPS forecasts with respect to the forthcoming announcement. In that case, the bimodality in the RND could reflect the divergence of beliefs between pessimistic and optimistic analysts with respect to the subsequently announced EPS. However, in unreported analysis, we do not find any systematic evidence of bimodality in histograms of EPS forecasts; the distributions of EPS forecasts are predominantly unimodal with a varying degree of dispersion and few outliers.

11 Recall that we compute RNDs only for the available range of traded strikes without extrapolating or fitting their tails. When we restrict our sample to RNDs that yield a cumulative probability of at least 70 percent, we find that 91 percent of the observations with CONCAVE = 1 exhibit a bimodal RND.

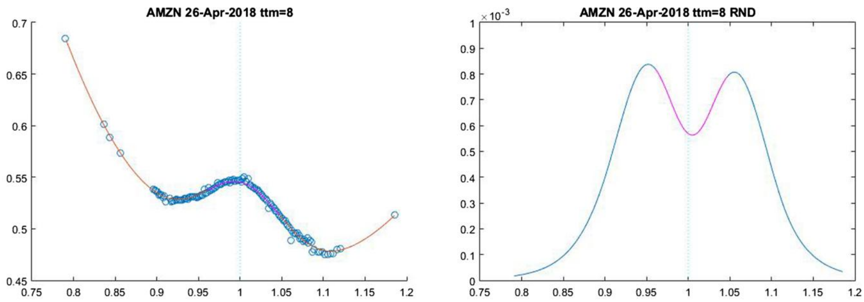

Figure 4. Concave IV curves and RND bimodality. This figure presents an example illustrating the correspondence between a concave IV curve and the RND for the underlying stock price. The left panel presents the IV curve for Amazon, computed from options with 8 days to expiry on April 26, 2018, that is, just before its quarterly earnings announcement. Circles indicate implied volatilities corresponding to actual traded strikes, whereas the curve is fitted using a quintic spline. The right panel presents the corresponding RND for Amazon on the same day. The RND is computed for the range of available strikes using the non-parametric methodology of Figlewski (2010).

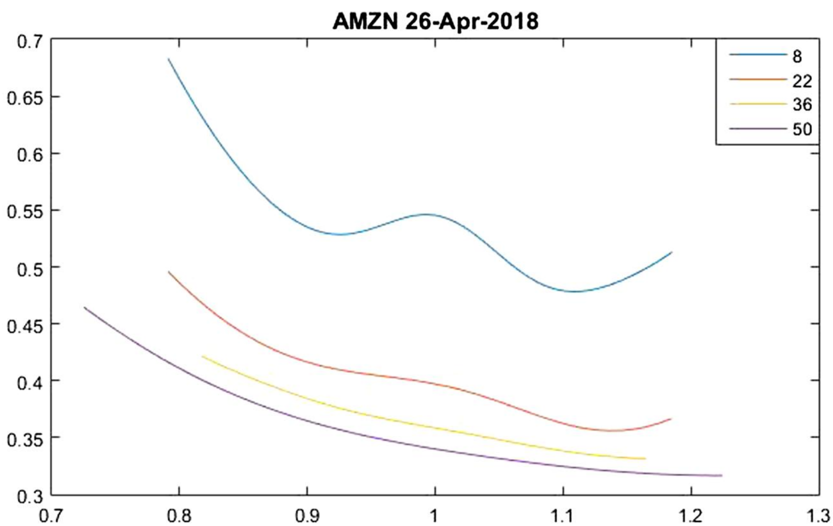

Figure 5. IV curves for short- versus longer-expiry options. This figure shows the shape of IV curves for Amazon, computed from options with different expiries (8, 22, 36, and 50 days to expiry) on April 26, 2018, that is, just before its quarterly earnings announcement. The IV curves are fitted using a quintic spline.

Intuitively, these patterns arise due to the relative effect of the anticipated stock price jump on EAD versus the diffusion component of the underlying process. As expiration shrinks, the effect of the anticipated price increase dominates the effect of the diffusion component, rendering the underlying RND bimodal and the IV curve concave. In contrast, as the time to expiration increases, the effect of the diffusion component dominates the effect of the anticipated price jump, the RND reverts to unimodality, and the IV curve becomes convex.

3.3 Bimodality in physical return distribution

We posit that the documented bimodality in the RND primarily reflects the sizeable probability that large stock price moves in either direction occur in EADs. In other words, we argue that prior to EAD, the distribution of short-term ahead stock returns can become bimodal under the physical measure $\mathbb{P}$, with modes away from 0 percent.[12]

An alternative hypothesis to explain RND bimodality is that option traders erroneously understate the probability of no news on EAD, and hence they systematically misprice options. This conjecture assumes that the probability distribution of stock returns on EAD is unimodal under $\mathbb{P}$ and that mispricing by option traders systematically renders the return distribution bimodal as we move from $\mathbb{P}$ to $\mathbb{Q}$.

To rule out this alternative hypothesis, we examine the distribution of stock returns on EADs. Figure 6 shows two such histograms. The top panel presents the histogram of realized stock returns on EADs following the formation of non-concave IV curves on $d-1$ ($\texttt{CONCAVE}=0$), whereas the bottom panel presents the corresponding histogram following the formation of concave IV curves on $d-1$ ($\texttt{CONCAVE}=1$). For both histograms, each bin has a 1 percent width, centered around the label on the x-axis. For example, the 0 percent bin contains observations with EAD returns between –0.5 percent and 0.5 percent. For each histogram, we also plot the corresponding density using an Epanechnikov kernel.

For $\texttt{CONCAVE}=0$, the top panel of figure 6 shows that the distribution of the realized stock returns on EADs exhibits unimodality in its central part, with the mode of the distribution located at the 0 percent bin. These features are similar to the ones found in the distribution of daily stock returns on typical trading days. The corresponding distribution when $\texttt{CONCAVE}=1$ in the bottom panel is strikingly different. We find that the central part of the distribution exhibits bimodality, with two distinct modes away from 0 percent. In fact, the left mode is located at the −2 percent bin and the right mode is located at the 2 percent bin. Moreover, the histogram shows that there are 9 bins with a frequency higher than the frequency of the 0 percent bin. Interestingly, for $\texttt{CONCAVE}=1$, EAD returns as high as around 4 percent or as low as around –4 percent are more likely to occur than returns around 0 percent.[13]

This evidence shows that the formation of bimodal RNDs prior to EADs when $\texttt{CONCAVE}=1$ does not simply derive from option traders systematically understating the probability of no news. To the contrary, these bimodal RNDs primarily arise because large stock returns in either direction are highly likely to occur on EADs and investors price these anticipated outcomes in the option market.

3.3.1 Concavity or high IV?

Since IV also increases in the run up to EADs (Patell and Wolfson 1979, 1981; DJKS 2019), could the bimodality in the distribution of EAD realized stock returns be actually driven by high IV per se rather than concavity? To check this, we sort the announcements according to their actual ATM IVs, computed from the corresponding short-expiry options on $d-1$, and select those with values greater than the median. We further divide this high IV sample into observations that exhibit concave IV curves ($\texttt{CONCAVE}=1$) and those that exhibit non-concave IV curves ($\texttt{CONCAVE}=0$) on $d-1$. This yields two subsamples of roughly similar sizes. We then compute the corresponding histograms of EAD realized stock returns for these two subsamples of high IV announcements.

Figure 7 shows these two histograms. Here, each bin has a 2 percent width, given the smaller subsamples. The top panel presents the histogram for the cases with high IV and $\texttt{CONCAVE}=0$, while the bottom panel presents the histogram for the cases with high IV and $\texttt{CONCAVE}=1$. We find that only the histogram computed from the concave observations exhibits the feature of bimodality in the central part of the EAD return distribution,

12 Of course, we do not argue that the return distributions under $\mathbb{P}$ and $\mathbb{Q}$ coincide.

13 These figures naturally mix high and low volatility firms. If we plot returns scaled by their recent realized volatility, we find patterns very similar to those reported in the figure. We conclude that the observed bimodality in the central part of the EAD return distribution following the formation of concave IV curves does not arise spuriously by potentially mixing high and low volatility firms.

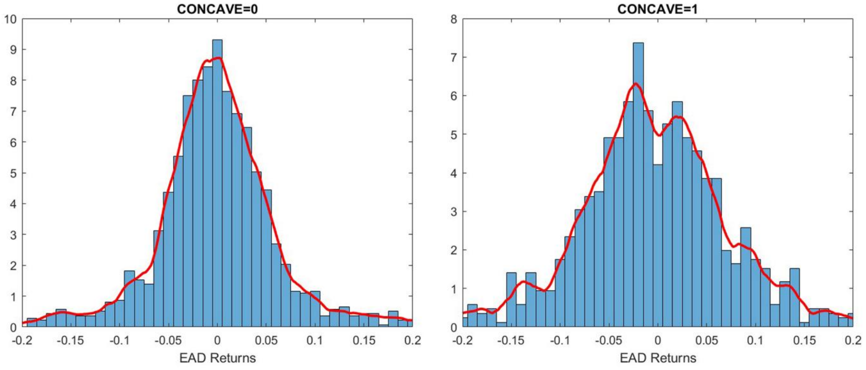

Figure 6. Distribution of realized stock returns on EAD following concave versus non-concave IV curves. This figure shows histograms of realized stock returns on EADs. The left panel illustrates the histogram for the observations associated with a non-concave (CONCAVE = 0) IV curve observed on the day prior to EAD. The right panel illustrates the corresponding histogram for the observations associated with a concave (CONCAVE = 1) IV curve. Each bin in the histogram has a 1 percent width, centered around the label on the x-axis. The corresponding density derived using an Epanechnikov kernel is also plotted in each panel. The sample consists of quarterly EADs during the period 2013–2020.

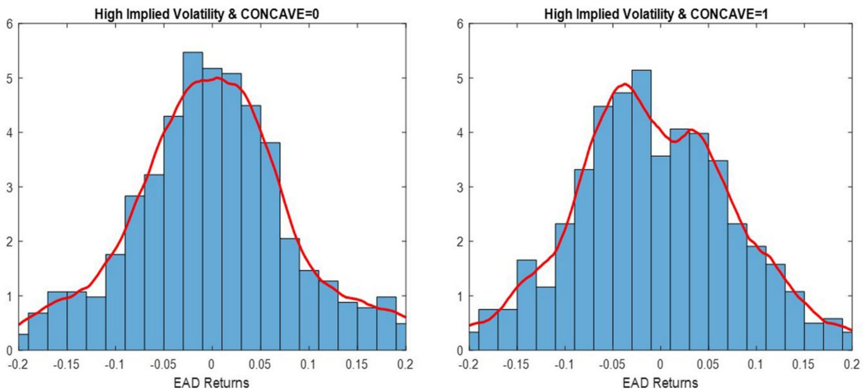

Figure 7. Distribution of realized stock returns on EAD for high IV announcements: Concave versus non-concave IV curves. This figure shows histograms of realized stock returns on EADs. Unlike figure 6, we consider only announcements characterized by higher than median IV values, computed from the corresponding ATM short-expiry options on the day prior to EAD. Each bin in the histogram has a 2 percent width, centered around the label on the x-axis. The corresponding density derived using an Epanechnikov kernel is also plotted in each panel. The sample consists of quarterly EADs during the period 2013–2020.

with both modes being away from 0 percent. In fact, for this subsample, we find that it is more likely to observe EAD returns as high as around 4 percent or as low as around –6 percent rather than returns around 0 percent. In contrast, the histogram for the non-concave observations reflects a disperse (as is to be expected, given the high IVs) but still unimodal distribution of EAD stock returns.[14]

[14] We repeat this analysis by alternatively sorting announcements based on their term structure of IV on $d-1$ (TSIV), which is an alternative empirical proxy for anticipated price jump volatility. We find again that it is the intersection of high TSIV and concave IV curve observations that yields a bimodal EAD stock return distribution, whereas the intersection of high TSIV and non-concave IV curve observations yields a disperse but unimodal EAD stock return distribution.

Thus, this evidence shows that high IV per se is not sufficient to yield a bimodal EAD stock return distribution. It is its interaction with concavity that yields this distinctive feature.

4. Model

Motivated by the empirical findings in the previous section, we build an option pricing model that can generate the novel features that we document, namely concavity in the IV curve and bimodality in the RND of the underlying stock price prior to the EAD. The model is based on the continuous-time model of Bates (1996), which features stochastic volatility and random jumps. We note that it is straightforward to generalize our modeling structure using more complicated continuous-time stochastic volatility models.

The most common modeling assumption for large stock price changes is that of a low-probability, randomly timed Poisson jump (see, e.g., Merton 1976; Ball and Torous 1985). Although random jumps, for example, can lead to an IV smirk and a left-tailed RND, capturing tail risk and explaining the expensiveness of OTM puts (see, e.g., Bates 1996, 2000; Pan 2002; Yan 2011), neither these jumps nor stochastic volatility typically generate concave IV curves.

DJKS (2019) and Piazzesi (2000) introduce deterministically timed jumps in the continuous-time path of the underlying stock price. Our model builds on their insights.[15] Let N_t^d count EADs prior to time t so that $N_t^d = \sum_j 1_{\tau_j \le t}$, where τ_j is an increasing sequence of predictable stopping times representing an EAD. Different from DJKS (2019), the jump size occurring on EAD τ_j, $Z_j = \ln(S_{\tau_j}/S_{\tau_j-})$, is assumed to follow a mixture of normal distributions. Specifically, $Z_j = \pi_j X_j^{(-)} + (1-\pi_j)X_j^{(+)}$, where π_j is a Bernoulli distribution with $P(\pi_j = 1) = p_j$ and $P(\pi_j = 0) = 1 - p_j$, $X_j^{(-)}|\mathcal{F}_{\tau_j-} \sim N(\mu_j^{(-)}, (\sigma_j^{(-)})^2)$ and $X_j^{(+)}|\mathcal{F}_{\tau_j-} \sim N(\mu_j^{(+)}, (\sigma_j^{(+)})^2)$.

This parsimonious modeling structure introduces event risk, which is captured by the anticipated upside and downside jumps, in the continuous-time stochastic volatility model. The model implies that on EAD, the stock price will exhibit either a "negative" jump $X_j^{(-)}$ with probability p_j or a "positive" one $X_j^{(+)}$ with probability $1 - p_j$. Since EADs are predictably timed, the martingale restriction (see Piazzesi (2000)) requires that $E[S_{\tau_j}|\mathcal{F}_{\tau_j-}] = S_{\tau_j-}$, imposing the following restriction upon the parameters of the Z_j distribution:

$$p_j \exp\left(\mu_j^{(-)} + 0.5\left(\sigma_j^{(-)}\right)^2\right) + (1-p_j)\exp\left(\mu_j^{(+)} + 0.5\left(\sigma_j^{(+)}\right)^2\right) = 1.$$

According to our model, under the risk-neutral measure $\mathbb{Q}$, the stock price and variance processes solve the following stochastic differential equations:

$$\begin{aligned} dS_t &= (r - \lambda_J \bar{\mu}_J) S_t dt + \sqrt{V_t} S_t dW_t^S + d\Big(\sum_{j=1}^{N_t^d} S_{\tau_j-}(e^{Z_j} - 1)\Big) + d\Big(\sum_{j=1}^{N_t} S_{\bar{\tau}_j-}(e^{\bar{Z}_j} - 1)\Big) \\ dV_t &= \kappa_v(\theta_v - V_t)dt + \sigma_v \sqrt{V_t} dW_t^v, \end{aligned} \tag{3}$$

where r is the risk-free rate, and W_t^S and W_t^v are two standard Brownian motions with correlation ρdt. Price jumps may also occur at random times $\bar{\tau}_j$ according to a Poisson process

[15] Todorov (2020) develops a nonparametric test that makes use of options with different expirations to detect the possibility of a jump in the underlying asset price at a given point in time and recover the corresponding jump distribution.

N_t with intensity parameter λ_J and jump size $\bar{Z}_j$ that is normally distributed, that is, $\bar{Z}_j|\mathcal{F}_{\bar{\tau}_j-} \sim N(\mu_J, \sigma_J^2)$; $\bar{\mu}_J$ denotes the random jump compensator given by $\bar{\mu}_J = \exp\left(\mu_J + \frac{1}{2}\sigma_J^2\right) - 1$. Finally, θ_v is the long-run mean of variance, κ_v determines the mean-reversion rate, and σ_v is the volatility-of-volatility parameter.

Note that if $p_j = 1$, the model collapses to that of DJKS (2019). However, in this case, the martingale restriction on Z_j implies that $\mu_j^{(-)} = -0.5\left(\sigma_j^{(-)}\right)^2$, and hence only a negative-mean jump can occur on EAD. Thus, our model is more general, allowing us to capture the impact of both upside and downside anticipated jumps on the stock price process. In the absence of EADs prior to time t (i.e., $N_t^d = 0$), the model collapses to that of Bates (1996).

We name our model SVJEJ2 (for stochastic volatility, Poisson jumps, and two expected jumps). The SVJEJ2 model specified in Equation (3) generates a conditional probability density function (pdf) for the log-return of the underlying stock that is a mixture of three constituent pdfs. The first one is derived by the diffusion and random jumps component of the model. The second and third pdfs are those of the two normally distributed upside and downside jumps that are anticipated to occur on EAD. The mixture of these three pdfs can generate a plethora of different distributions. These include asymmetric distributions, distributions with fat tails, and most importantly, for our analysis, multi-modal distributions. To this end, this parametric model is sufficiently flexible to reproduce RNDs that prior studies have empirically recovered from option prices by fitting mixtures of log-normal distributions or smoothing splines in the IV space (see, e.g., Leahy and Thomas 1996; Melick and Thomas 1997; Birru and Figlewski 2012; Mirkov, Pozdeev, and Söderlind 2016; Hanke, Poulson, and Weissensteiner 2018).

Our proposed SVJEJ2 model captures the increase in IV in the run-up to the earnings announcement and its sharp fall right after (Patell and Wolfson 1979, 1981). In addition, similar to DJKS (2019), the SVJEJ2 model can generate a downward sloping IV term structure prior to scheduled announcements due to the anticipated price jump. However, unlike the prior literature, the SVJEJ2 model can also generate concave IV curves and bimodal RNDs. Hence, it disentangles the effect of a scheduled event on the overall level of IV from the corresponding effects across different moneyness. As a result, the SVJEJ2 model can provide estimates of the probability, direction, expected magnitude, and dispersion of price jumps that are anticipated to occur. This rich set of information can help us infer investor expectations regarding the impending event, as well as the pricing of the arising event risk.

We emphasize that we do not claim that our SVJEJ2 model is the only one that can generate a bimodal RND. For example, in the case of the classic jump-diffusion model, a sufficiently high probability of a jump whose size exhibits a large mean but relatively low variance can give rise to a mode in the tail of the distribution, leading to bimodality. However, our focus is on bimodality in *central part* of the RND and concavity in the IV curve; introducing both upside and downside anticipated jumps, our SVJEJ2 model can more naturally capture these phenomena.

In the SVJEJ2 model, the stock price is defined by the product of an affine component and a discrete jump on EADs. Therefore, option pricing proceeds in a similar fashion to standard affine models, using the closed-form solution of the conditional characteristic function of the log stock price (see Bates 1996; Duffie, Pan, and Singleton 2000). The characteristic function is presented in Appendix B.

To showcase the ability of the SVJEJ2 model to generate concave IV curves and bimodal RNDs prior to EADs, we consider the example of Apple on October 28, 2013. Recall that in this case, a single EAD occurs prior to option expiry. The corresponding empirical IV curve derived from the fitting of a quintic spline to the actual IVs is illustrated in Panel A of figure 2. Here, we fit our model to the actual option prices by minimizing the root mean squared error (RMSE) between the actual and model-implied IVs. The parameter values that minimize the RMSE are: $\theta_v = 0.42$, $\kappa_v = 2.36$, $\sigma_v = 4.67$, $\rho = -0.01$, $\lambda_J = 14.86$, $\mu_J = -0.007$, $\sigma_J = 0.078$, $p_j = 0.444$, $\mu_j^{(-)} = -0.056$, $\sigma_j^{(-)} = 0.005$, $\mu_j^{(+)} = 0.043$, $\sigma_j^{(+)} = 0.006$. Apart from pronounced stochastic volatility and a random price jump whose size has a small mean but is rather disperse, these parameter values indicate that investors anticipate with risk-neutral probability 44.4 percent (55.6 percent), a downside (upside) price jump on EAD with mean size -5.6 percent (4.3 percent) and low volatility.

Figure 8 shows that these parameter values generate a W-shape IV curve that fits the actual IVs extremely well. We repeat the same process, fitting the DJKS (2019) model to the actual option prices. As illustrated in figure 8, the latter model yields a poor fit because it cannot generate concavity in the IV curve.

Figure 9 presents the RND for Apple's log stock return on October 28, 2013. This RND is computed using the parameter values estimated above. It is evident that our SVJEJ2 model gives rise to a bimodal RND, which is very similar to the empirical RND corresponding to the fitted IV curve using the quintic spline.

4.1 Model fit

We formally compare the goodness of fit of our SVJEJ2 model to that of DJKS (2019) model. We fit both models for each firm-announcement observation in our sample with at least twelve strikes satisfying the data filters specified in Section 2.1. This requirement ensures the reliability of the fitted parameter values and yields a set of 1,373 firm-announcements. We fit both models to the actual prices of the short-expiry options on $d-1$ by minimizing the RMSE between the actual and model-implied IVs.

Table 4 reports statistics for the in-sample fit of these two models. We report these statistics for all firm-announcements in the estimation sample, for subsamples with concave (`CONCAVE` $= 1$) or non-concave (`CONCAVE` $= 0$) IV curves observed on day $d-1$, and for subsamples with higher or lower than the median ATM IV value, calculated from the corresponding short-expiry options on $d-1$.

Column (1) of Table 4 shows that the average RMSE of SVJEJ2, at 0.44 percent, is much lower than that of the DJKS model, at 0.97 percent, across the full sample. The superior performance of SVJEJ2 is much more pronounced for high IV announcements, where its RMSE is less than half of the RMSE of the DJKS model (0.67 percent versus 1.49 percent), as well as for concave IV curves, where its RMSE is less than a third of that of the DJKS model (0.44 percent versus 1.41 percent). Importantly, these results also reveal the inferior performance of DJKS for the intersection of high IV and concave observations, where its RMSE is four times higher than that of the SVJEJ2 model (0.45 percent versus 1.97 percent).

Motivated by the illustrative example of Apple in figure 8, we further examine how well each model can fit the actual ATM IV. Specifically, we measure the potential bias of each model as the difference between the model-implied and the actual ATM IV. Column (2) of Table 4 shows that the DJKS model features a downward bias, as it underestimates ATM IV by 0.78 percent. This negative bias is much larger for observations with high ATM IV values (-1.31 percent) and concave IV curves (-1.30 percent). For the intersection of high ATM IV and concave observations, the DJKS model yields a negative bias of -1.93 percent. In contrast, our SVJEJ2 model precisely fits the actual ATM IV, exhibiting an overall negligible bias (-0.01 percent). Even in the case of high ATM IV and concave observations, the corresponding bias is extremely small (-0.08 percent).

To understand the source of this downward bias for the DJKS model, we compute for each model the component of (risk-neutral) volatility due to the anticipated price jump on EAD, denoted as `EADJumpVol`. For the SVJEJ2 model, this is given by

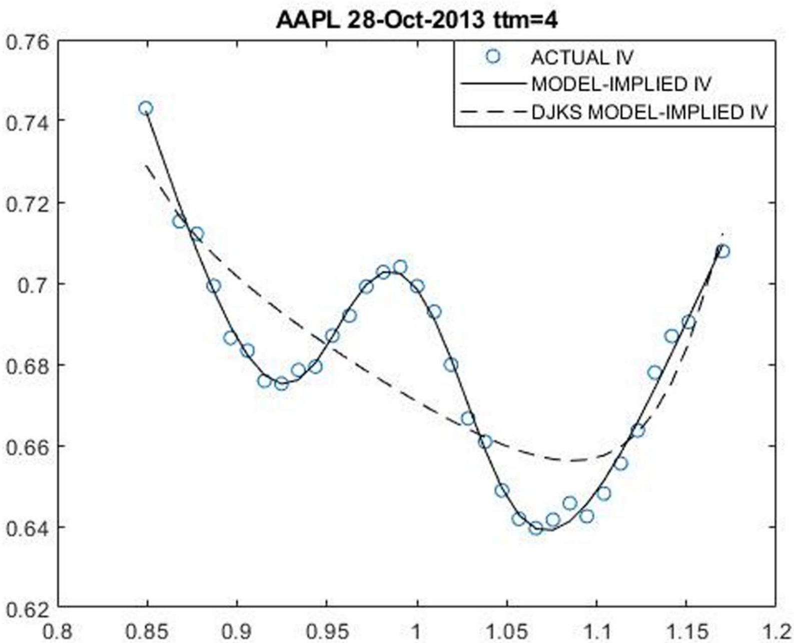

Figure 8. Model-implied IV curve. This figure shows the fit of the model-implied IV curve (solid curve) relative to the actual IVs (circles). Actual IVs are computed for Apple from options with 4 days to expiry on October 28, 2013, that is, prior to its quarterly earnings announcement. The model-implied IV curve is computed using the corresponding estimated parameter values for the model specified in Section 4. Model parameter values are estimated by minimizing the RMSE between the actual and the model-implied IVs. The figure also shows the corresponding IV curve (dashed curve) implied by fitting the DJKS (2019) model to the actual IVs.

$EADJumpVol_{SVJEJ2} = \sqrt{\frac{1}{T}\left(p_j^2\left(\sigma_j^{(-)}\right)^2 + (1-p_j)^2\left(\sigma_j^{(+)}\right)^2\right)}$, where T is the maturity of the option. For the DJKS model, this component is given by setting $p_j = 1$ in the above expression, that is, $EADJumpVol_{DJKS} = \sqrt{\frac{1}{T}\left(\sigma_j^{(-)}\right)^2}$.

Column (3) of Table 4 shows the corresponding annualized average values for `EADJumpVol`. Naturally, the estimates of `EADJumpVol` are much higher for high IV announcements as well as for announcements with concave IV curves. We find that the DJKS model yields a substantially smaller magnitude of `EADJumpVol` relative to the SVJEJ2 model across all subsamples (15.69 percent versus 26.35 percent for the full sample). This happens because in the DJKS model, `EADJumpVol` is driven by the variance of a single negative-mean jump. In contrast, the SVJEJ2 model relaxes this restriction and allows `EADJumpVol` to be driven by the variance of both upside and downside jumps. Hence, we conclude that a single anticipated price jump cannot fully capture the uncertainty priced in the option market prior to EADs. This misspecification leads the DJKS model to underestimate the anticipated price jump volatility component and, in turn, leads to a downward bias with respect to the actual ATM IV.

Finally, we examine the implications of each model with respect to the term structure of IV. Recall that we compute `TSIV` using the actual ATM IV from the short-expiry option (T_1) and the corresponding ATM IV from the next available expiration (T_2). We can alternatively compute this term structure variable using model-implied ATM IVs for the corresponding maturities. Specifically, as described above, for each firm-announcement, we fit

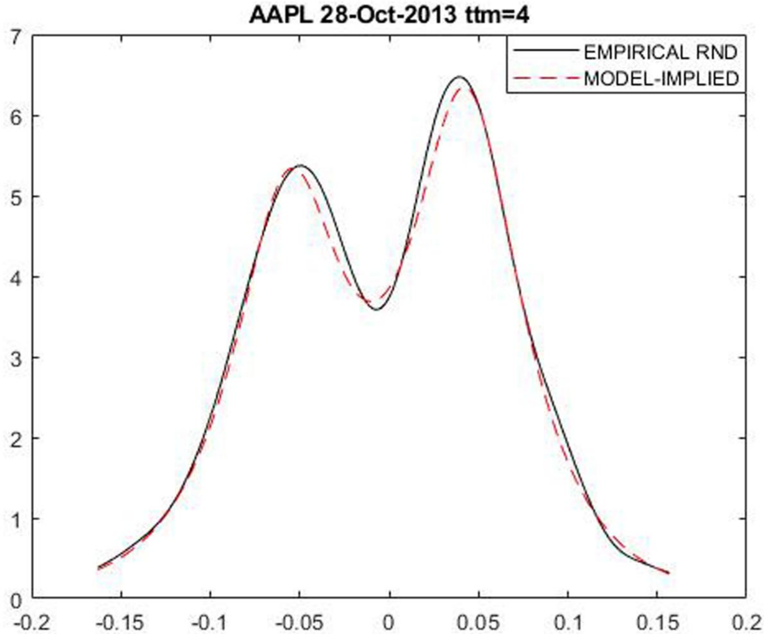

Figure 9. Model-implied RND. This figure shows the empirical RND (solid curve) together with the model-implied RND (dashed curve) for Apple's log stock return, derived from options with 4 days to expiry on October 28, 2013, that is, prior to its quarterly earnings announcement. The empirical RND corresponds to the empirical IV curve, which is derived from fitting a quintic spline to the actual IVs. The model-implied RND is computed using the corresponding estimated parameter values for the model specified in Section 4. Model parameter values are estimated by minimizing the RMSE between the actual and the model-implied IVs.

each of the two models on $d-1$ to the actual prices of the short-expiry options with maturity T_1. We then use these fitted parameter values to compute the model-implied ATM IV for maturities T_1 and T_2, which in turn yield the model-implied `TSIV`. Comparing the model-implied and actual `TSIV`, we examine the ability of each model to capture the actual term structure of IV.

Column (4) of Table 4 shows that the DJKS model substantially underestimates the actual `TSIV` (the average bias across all observations is -5.16 percent), yielding a flatter than observed term structure. Again, the downward bias of the DJKS model is larger for announcements with high IV values (-6.17 percent) and concave IV curves (-5.70 percent) on $d-1$. The magnitude of this downward bias is even larger (in absolute value) than the corresponding bias of the DJKS model with respect to the short-expiry ATM IV. These results suggest that the underestimation of `EADJumpVol` by the DJKS model leads to an overestimation of the next-expiry ATM IV, and hence to a much lower `TSIV` estimate. In contrast, the SVJEJ2 model exhibits a much smaller downward bias (-1.44 percent in the full sample), whose magnitude is less than a third of the corresponding bias of the DJKS model across all subsamples considered. Hence, apart from accurately fitting the ATM IV of the short-expiry (T_1) option, the SVJEJ2 model also yields a much more accurate estimate of the ATM IV for maturity T_2.

The analysis in this section leads to two main conclusions. First, the SVJEJ2 model can generate the distinctive feature of concavity in the IV curve, which is in turn associated with the bimodality in the distribution of the EAD realized stock returns. Second, the SVJEJ2 model can more accurately fit the actual ATM IV and `TSIV`. Hence, it can better capture the informational content of short-expiry options prior to EAD. To the contrary,

Table 4. Model fit.

This table reports statistics related to the goodness of fit for the DJKS model (Panel A) and the SVJEJ2 model (Panel B). Both models are fitted to the prices of short-expiry options on $d-1$, where d is the EAD, for each firm-announcement in our sample with at least twelve strikes. The fitted model parameter values minimize the RMSE between the actual and the model-implied IVs. The average values of these statistics are reported for all announcements, for subsamples with concave (CONCAVE = 1) or non-concave (CONCAVE = 0) IV curves observed on $d-1$, and for subsamples with higher or lower than the median actual ATM IV value on $d-1$, computed from the corresponding short-expiry options. Bias ATM IV is calculated as the model-implied minus the actual ATM IV on $d-1$. EADJumpVol denotes the component of risk-neutral volatility on $d-1$ due to the anticipated price jump on EAD, based on the corresponding fitted parameter values for each model. Bias TSIV is calculated as the model-implied minus the actual TSIV measured on $d-1$. All numbers are in percent. The sample consists of quarterly earnings announcements during the period 2013–2020.

	RMSE (1)	Bias ATM IV (2)	EADJumpVol (3)	Bias TSIV (4)
Panel A: DJKS model				
			All observations	
All	0.97	−0.78	15.69	−5.16
High ATM IV obs.	1.49	−1.31	27.04	−6.17
Low ATM IV obs.	0.45	−0.26	4.31	−4.16
			CONCAVE = 1	
All	1.41	−1.30	20.22	−5.70
High ATM IV	1.97	−1.93	36.48	−6.02
Low ATM IV	0.86	−0.67	3.95	−5.39
			CONCAVE = 0	
All	0.57	−0.32	11.66	−4.54
High ATM IV	0.87	−0.51	18.79	−5.74
Low ATM IV	0.27	−0.13	4.51	−3.33
Panel B: SVJEJ2 model				
			All observations	
All	0.44	−0.01	26.35	−1.44
High ATM IV	0.67	−0.05	37.66	−1.76
Low ATM IV	0.21	0.01	15.03	−1.12
			CONCAVE = 1	
All	0.44	−0.03	27.50	−1.40
High ATM IV	0.45	−0.08	40.36	−1.71
Low ATM IV	0.44	0.03	14.60	−1.10
			CONCAVE = 0	
All	0.43	−0.01	25.32	−1.48
High ATM IV	0.46	−0.02	36.29	−1.81
Low ATM IV	0.40	0.00	14.31	−1.15

the single-jump model of DJKS substantially underestimates the magnitude of the anticipated price jump volatility, and hence it yields downward biased estimates of the ATM IV and TSIV, especially in the presence of concave IV curves.

5. Pricing information of concave IV curves

We now turn our focus on the informational content of concave IV curves with respect to the pricing of event risk. To this end, this section considers alternative option strategies that capture this aspect of risk and examines how CONCAVE is related to the corresponding premium. To motivate this analysis, we first provide further evidence to support the proposition that concavity in the IV curve prior to EADs is a valid ex-ante signal for event risk.

5.1 Do concave IV curves signal event risk?

According to the definition of Liu, Longstaff, and Pan (2003), the realization of event risk would lead to a large shock to stock prices and volatility. Therefore, we examine whether concave IV curves are associated with higher absolute stock returns on EAD relative to non-concave IV curves. To ensure that our results are not driven by market-wide price movements or systematic factor-related returns, we use the absolute abnormal stock return on EAD (ABSEADABRET) with respect to the Fama–French–Carhart (FFC) four-factor model (see the definition in Appendix A).

The summary statistics reported in Table 1 show that the average ABSEADABRET is 4.86 percent, whereas the median is 3.41 percent. These statistics are consistent with the finding in the prior literature that stock prices often exhibit very large movements around earnings announcements (see Lee and Mykland 2008; Lee 2012; Kapadia and Zekhnini 2019). More importantly, Table 3 shows that the average ABSEADABRET is 5.88 percent when CONCAVE = 1 and 4.24 percent when CONCAVE = 0. The differential return of 1.64 percent is strongly significant (t-statistic = 7.71). Panel A of Supplementary Appendix Table IA1 examines this relationship using panel regressions and controlling for a number of firm characteristics. These results confirm that, on average, concave IV curves are followed by a significantly higher absolute abnormal stock return on EAD relative to non-concave IV curves.

To further show that the above differential in absolute stock returns does not simply reflect differences in the level of stock volatilities, we calculate the ratio of ABSEADABRET to idiosyncratic volatility, which is estimated from $d-250$ to $d-25$ using the FFC four-factor model. Such ratios have been used in prior literature as measures of stock price jumps (see, e.g., Kapadia and Zekhnini 2019). We find that the average ratio of ABSEADABRET to idiosyncratic volatility is equal to 3.35 following the formation of concave IV curves and 3.07 following non-concave IV curves, with the difference being significant (t-statistic = 2.44). Hence, we confirm that concave IV curves genuinely signal stock price jumps on EADs.

Next, we examine whether CONCAVE is also related to the post-EAD stock return volatility (POSTEADVOL). In our benchmark analysis, we calculate the annualized 10-day stock return volatility from d to $d+9$ using the standard formula provided in Appendix A. Even though POSTEADVOL is naturally affected by the magnitude of ABSEADABRET, it also captures stock price fluctuations occurring after the EAD.

The mean (median) POSTEADVOL reported in Table 1 is 41.09 percent (33.39 percent). Notably, Table 3 shows that the average POSTEADVOL is 47.49 percent when CONCAVE = 1, whereas the corresponding average value for CONCAVE = 0 is 37.11 percent, with the difference of 10.38 percent being highly significant (t-statistic = 8.82). Panel B of Supplementary Appendix Table IA1 examines this relationship using panel regressions. We similarly find that concave IV curves are followed by significantly higher POSTEADVOL, compared to nonconcave IV curves. Hence, we confirm that concavity in the IV curve prior to EAD also signals much higher post-announcement stock volatility.

5.2 Straddle returns

Having established that concavity in the IV curve is a valid ex-ante signal of event risk, we first examine how CONCAVE is related to straddle returns on EADs. Anticipating large stock price swings around EADs, investors could take long positions in ATM straddles to speculate or hedge against these swings regardless of their direction.

We compute the returns of delta-neutral ATM straddles (STRADDLE) on EAD, using the nearest-to-the-money pair of call and put options within the moneyness (K/S) range of 0.98 to 1.02. We buy the straddle at the close of the trading day prior to the EAD and sell it at the close after the announcement. We use the shortest available options, with expiry between 4 and 13 days. The return of the delta-neutral straddle on EAD is given by:

$$\texttt{STRADDLE} = wR_c + (1 - w)R_p, \tag{4}$$

where R_c (R_p) is the return of the call (put) option on EAD. The weight w is given by:

$$w = -\frac{\Delta_P/P}{\Delta_C/C - \Delta_P/P}, \tag{5}$$

where Δ_C (Δ_P) is the delta of the call (put) provided by OptionMetrics and C (P) is the corresponding call (put) price at straddle formation.

The summary statistics reported in Table 1 show that the median STRADDLE on EAD is -15.43 percent. This finding supports the argument that investors most often pay a substantial price to be hedged against the increased volatility and large stock price swings observed around EADs. However, STRADDLE exhibits a positively skewed distribution in our sample and its average is only -0.86 percent. Interestingly, Table 3 shows that the average STRADDLE return is -3.74 percent when CONCAVE = 1 and 0.91 percent when CONCAVE = 0, with a differential return of –4.65 percent (t-statistic = –2.16). This evidence suggests that investors pay a significant premium to hedge against the large price swings that arise due to earnings announcements only when IV curves become concave.

Table 5 presents estimates from predictive panel regressions of STRADDLE on CONCAVE as well as a number of controls also measured on the day prior to the EAD.[16] Across all specifications, we control for the expiry and the average moneyness of the pair of options used to construct this straddle strategy, ensuring that our results are not driven by these features. Column (1) shows that concave IV curves are followed by a 4.57 percent (t-statistic = 2.40) lower average straddle return, as compared to non-concave IV curves.[17] In column (2), we control for firm characteristics BETA, Ln(SIZE), B/M, RUNUP, MOM, Ln(PRICE), DISP, and ANNBETA (the cofficients on these controls are suppressed for brevity). We find that the coefficient on CONCAVE increases (in absolute value) to 7.60 percent and is strongly statistically significant.

As mentioned before, the literature has established that ATMIV also increases, and hence RVIV becomes substantially more negative, in the run up to EADs (Patell and Wolfson 1979, 1981). In addition, DJKS (2019) document that TSIV takes highly positive values prior to earnings announcements. In fact, Table 3 shows that concave IV curves are typically associated with higher ATMIV and TSIV values and more negative RVIV values. Therefore, these three option-based variables could serve as alternative proxies for event/jump risk, with TSIV explicitly suggested as such by DJKS (2019). As a result, one might wonder whether the reported negative straddle premium is just a manifestation of increased short-term volatility prior to EADs per se, as captured by ATMIV, RVIV, and TSIV, rather than of concavity.

16 We have repeated the analysis reported in Table 5 using simple instead of delta-neutral ATM straddle returns. The results, available upon request, are very similar to the ones reported in Table 5.

17 This differential straddle return is computed using mid-point option prices. To alleviate the potential concern that this might be driven by options' bid–ask spread, we alternatively compute straddle returns taking this spread into account. In the most stringent version, we compute returns using ask prices to buy the straddle at $d-1$ and bid prices to sell the straddle at d. Alternatively, in the spirit of Muravyev and Pearson (2020), who argue that the effective option trading costs are lower than what the conventionally quoted bid–ask spreads indicate, we also compute straddle returns using adjusted bid–ask spreads. Specifically, we assume that the effective bid–ask spread on a given day is 75 percent or 50 percent, respectively, of the quoted spread. Using these effective bid–ask spreads, we compute the corresponding bid and ask prices. Naturally, the straddle returns are overall substantially lower when we take transaction costs into account. Nevertheless, the differential straddle return on EAD following concave versus non-concave IV curves remains intact and significant, similar to the one reported in Table 5. Hence, we conclude that the sign and the magnitude of this return differential is not affected by options' bid–ask spread.

Table 5. Concave IV curves and delta-neutral straddle returns on EAD.

This table presents results from predictive panel regressions of delta-neutral ATM straddle returns (STRADDLE) computed on EAD on CONCAVE and other variables. Stock controls, measured on the day prior to the EAD, in column (2) include BETA, Ln(SIZE), B/M, RUNUP, MOM, Ln(PRICE), DISP, and ANNBETA. The definition of all the variables is provided in Appendix A. All specifications include an intercept and two option controls, viz. the expiry and the average moneyness of the options used to construct the straddle strategy. Coefficient estimates on ATMIV, RVIV, and TSIV are multiplied by 100 for easy readability. *t*-statistics in parentheses are calculated using two-way clustered standard errors at the firm- and quarter-level. The sample consists of quarterly earnings announcements during the period 2013–2020.

	(1)	(2)	(3)	(4)	(5)
CONCAVE	−4.57	−7.60	−5.27	−6.77	−6.31
	(−2.40)	(−3.49)	(−2.78)	(−3.12)	(−3.10)
ATMIV	—	—	−0.14	−26.69	0.06
			(−0.03)	(−2.38)	(0.01)
RVIV	—	—	—	−16.16	—
				(−2.77)	
TSIV	—	—	—	133.86	—
				(2.06)	
RNS	—	—	—	—	−6.03
					(−1.72)
RNK	—	—	—	—	−5.11
					(−3.97)
Stock Controls	No	Yes	No	No	No
# observations	2,175	1,930	2,151	2,139	2,151
R^2	0.27	1.34	0.32	0.74	0.78

To address this question, we include ATMIV, RVIV, and TSIV as control variables in the predictive regressions. To avoid overfitting and concerns of multicollinearity, we omit firm controls is these specifications, but results are very similar if we do include them. Columns (3) and (4) of Table 5 show that the magnitude and significance of the coefficient on CONCAVE remain intact—even increase slightly—in the presence of these three variables.

Our study posits that concavity in the IV curve is typically associated with bimodality in the central part of the RND. Hence, we further examine whether the informational content of CONCAVE with respect to the pricing of event risk is distinct from the one contained in the tails of the RND. To this end, we include RNS and RNK as control variables in the predictive regressions. By definition, RNS and RNK capture the tails of the RND and could proxy for tail risk. The estimated coefficients in column (5) of Table 5 show that the coefficient on CONCAVE remains highly significant in the presence of these risk-neutral moments.

The main conclusion from this analysis is that when IV curves become concave, investors pay a substantial premium to hedge against the larger than average stock price swings that are typically observed on these EADs. In fact, even though larger than average stock price movements do occur on EADs following the formation of concave IV curves, these price swings are not large enough to offset the substantial cost of purchasing straddles on these occasions.

To provide direct evidence that ATM straddles are particularly costly in the presence of concave IV curves, we introduce an intuitive measure of their expensiveness. Specifically, we calculate the following ratio:

$$\mathtt{IMPMOVE} = \frac{C+P}{S}, \tag{6}$$

where, as above, C (P) is the ATM call (put) price at straddle formation, that is, on the day prior to EAD, and S is the corresponding price of the underlying stock. This measure roughly indicates how much the underlying stock price should move in either direction to offset the cost of a symmetric ATM straddle, and hence it is termed as the implied stock price move (IMPMOVE). The higher (lower) the IMPMOVE is, the more (less) expensive it is to purchase an ATM straddle, ceteris paribus. Supplementary Appendix Table IA2 presents estimates from contemporaneous panel regressions of IMPMOVE on CONCAVE and a number of firm characteristics also measured on the day prior to EAD. We find that concave IV curves are associated with a 2.21 percent higher IMPMOVE relative to non-concave IV curves.

The significantly higher cost of buying straddles in the presence of concave IV curves provides an alternative way to illustrate that investors pay a significantly higher price to hedge against the event risk that arises on these occasions due to the impending announcement. This corroborates the argument that concave IV curves provide an ex-ante signal of event risk. Based on these findings, we conclude that investors can ex-ante identify the announcements that trigger large stock price moves and they pay a substantially higher premium to hedge against them, most obviously by purchasing straddles.[18] As a result of this hedging activity, ATM options become very expensive, trading at higher volatility, and hence the corresponding IV curves turn concave prior to EADs.

5.3 Gamma or vega risk?

Delta-neutral straddle returns have been often used to capture the price of volatility risk (see, e.g., Coval and Shumway 2001). However, this interpretation holds true only for small diffusive shocks. In the presence of jumps, delta-neutral straddles expose investors to both stochastic volatility (vega) and jump (gamma) risk. Thus, we cannot distinguish between these two sources of risk using these straddle returns (Cremers, Halling, and Weinbaum 2015). This is particularly important around EADs as stock prices often jump upon the announcement.

We disentangle these two effects in this section by examining whether gamma or vega risk is priced around earnings announcements, and by analyzing the source of risk concave IV curves are related to. We follow two complementary approaches. First, similar to Dew-Becker, Giglio, and Kelly (2021), we construct strangles and compute their returns around EADs. Strangles yield positive payoffs only when the underlying price exhibits a sufficiently large move. Hence, strangle returns can provide direct evidence on the price of gamma risk around EADs. Second, following Cremers, Halling, and Weinbaum (2015), we compute the EAD returns of delta- and vega-neutral ATM straddles, which expose investors to gamma risk only, as well as the corresponding returns of delta- and gamma-neutral ATM straddles, which expose investors to vega risk only. Since these calendar straddle strategies can isolate each dimension of risk, their returns can be directly attributed to gamma and vega risk exposure, respectively.

5.3.1 Strangle returns

Different from straddles, a strangle is a portfolio of long positions in an OTM call and an OTM put. Therefore, its payoff is typically negative unless a sufficiently large movement in the price of the underlying asset occurs. We form delta-neutral strangles at the end of the trading day prior to EAD and unwind the position at the close of EAD. As with straddles, we use the shortest available options, with expiry between 4 and 13 days. Similar to Dew-Becker, Giglio, and Kelly (2021), we use strikes that are nearest to one standard deviation

[18] A risk-based explanation of our results is consistent with the prior literature that has predominantly interpreted negative delta-neutral straddle returns as the premium investors pay to hedge against volatility and/or jump risk (see, e.g., Coval and Shumway 2001; Cremers, Halling and Weinbaum 2015; DJKS 2019). de Silva, So, and Smith (2023) propose an alternative behavioral interpretation, namely that negative straddle returns result from retail investors, who are attracted by earnings announcements with anticipated spikes in volatility and irrationally overpay for ATM options. For our sample of very large capitalization firms with very liquid options, a purely irrational explanation is less likely to hold. Moreover, we observe negative average straddle returns *only* in the presence of concave IV curves, which are shown to capture jump risk in the underlying stock price.

(scaled by time to expiry) away from the underlying stock price at formation.[19] We require that the absolute difference between the available (i.e., traded) and the desired moneyness (K/S) for the OTM options does not exceed 0.01, but we typically have a very close match. The STRANGLE return on EAD is computed in a similar fashion to equation (4), with the relevant weights assigned to OTM options ensuring that the strangle is delta-neutral at formation.

Table 1 reports that the median STRANGLE return on EAD is -28.78 percent. This is almost twice as large as the median STRADDLE return and indicates that investors most often pay a substantial premium to hedge against the gamma risk that arises due to earnings announcements. As expected, STRANGLE exhibits a positively skewed distribution and its average is -2.32 percent, revealing a negative price for gamma risk around EADs. More interestingly, Table 3 shows that the average STRANGLE return is -7.94 percent when CONCAVE $=1$ and 1.12 percent when CONCAVE $=0$, with differential return of -9.05 percent (t-statistic $=-2.45$). Hence, investors pay a substantial premium to hedge against gamma risk only when IV curves become concave prior to the EAD. In fact, the cost of purchasing a strangle in the presence of a concave IV curve is so high, on average, that it cannot be offset by the large stock returns that are often observed on EADs.

Table 6 confirms this finding in a predictive panel regression setup. Here, we also control for the expiry of the options used to construct the strangle and the absolute difference in the moneyness levels between the available and the desired strikes, to ensure that the reported return differential is not driven by these features. Column (1) shows that concave IV curves are followed by a 8.84 percent (t-statistic $=-2.41$) lower average strangle return relative to non-concave IV curves. Columns (2)–(5) show that this differential becomes even greater as we control for other firm characteristics, or option-based risk measures, such as ATMIV, RVIV, TSIV, RNS, and RNK. In sum, we find strong evidence that in the presence of concave IV curves, investors pay a substantial premium to hedge against the large stock price moves observed on EADs.

5.3.2 Delta- and vega-neutral straddle returns

An alternative way to isolate the effects of vega and gamma risk is to examine calendar straddles that combine a short- and a long-maturity delta-neutral ATM straddle.[20] First, we construct a delta- and vega-neutral ATM straddle (JUMPSTRADDLE), which exposes investors to gamma risk only. We form this strategy at $d-1$ and unwind it at d. The strategy consists of two legs. The first leg is a long position in a delta-neutral straddle constructed from the nearest-to-the-money options with the shortest available expiry between 4 and 13 days. The second leg is a short position in $\mathcal{V}_S/\mathcal{V}_L$ delta-neutral straddles using the nearest-to-the-money options with the longest available expiry between 100 and 180 days, where $\mathcal{V}_S$ ($\mathcal{V}_L$) denotes the vega of the short- (long-) maturity straddle.[21] This position ensures that this calendar strategy is vega-neutral at formation. The JUMPSTRADDLE return on EAD is given by:

$$\text{JUMPSTRADDLE} = wR_S + (1-w)R_L, \tag{7}$$

where R_S (R_L) is the return of the shorter- (longer-) maturity delta-neutral straddle on EAD and $w = -(\mathcal{V}_L/V_L)/(\mathcal{V}_S/V_S - \mathcal{V}_L/V_L)$, with V_S (V_L) denoting the value (i.e., cost) of the shorter- (longer-) maturity straddle at formation.

19 We use the 30-day realized volatility that is available at $d-1$ from the Historical Volatility File of OptionMetrics.

20 In a recent study, Chen, Gan, and Vasquez (2023) use this approach to decompose the price of a straddle into its volatility and jump components. They find that the behavior of straddle returns around earnings announcements is mostly driven by the jump component. Our results are consistent with this finding but we further show that gamma risk carries a significant premium prior to EADs only in the presence of concave IV curves.

21 We require the moneyness of the utilized calls and puts to lie within the range 0.98–1.02 for the short-maturity straddle and 0.96 to 1.04 for the longer-maturity straddle, but it is typically very close to 1.

Table 6. Concave IV curves and delta-neutral strangle returns on EAD.

This table presents results from predictive panel regressions of delta-neutral strangle returns (STRANGLE) computed on EAD on CONCAVE and other variables. Stock controls, measured on the day prior to the EAD, in column (2) include BETA, Ln(SIZE), B/M, RUNUP, MOM, Ln(PRICE), DISP, and ANNBETA. The definition of all the variables is provided in Appendix A. All specifications include an intercept and two option controls, viz. the expiry and the average moneyness of the options used to construct the stangle strategy. Coefficient estimates on ATMIV, RVIV, and TSIV are multiplied by 100 for easy readability. *t*-statistics in parentheses are calculated using two-way clustered standard errors at the firm- and quarter-level. The sample consists of quarterly earnings announcements during the period 2013–2020.

	(1)	(2)	(3)	(4)	(5)
CONCAVE	−8.84	−12.92	−10.06	−11.83	−11.93
	(−2.41)	(−3.29)	(−2.70)	(−2.91)	(−2.85)
ATMIV	—	—	2.72	−42.65	2.91
			(0.28)	(−1.91)	(0.30)
RVIV	—	—	—	−38.74	—
				(−2.46)	
TSIV	—	—	—	182.88	—
				(1.45)	
RNS	—	—	—	—	−11.12
					(−2.06)
RNK	—	—	—	—	−9.29
					(−3.27)
Stock Controls	No	Yes	No	No	No
# observations	1,903	1,710	1,883	1,874	1,883
R^2	0.4	1.95	0.5	0.92	1.04

Table 1 shows that the average JUMPSTRADDLE return on EAD is −2.82 percent, indicating again a negative price for gamma risk. Table 3 further shows that this negative premium accrues from observations with a concave IV curve. The average JUMPSTRADDLE return is −10.69 percent when CONCAVE = 1 and 2.29 percent when CONCAVE = 0, yielding a significant differential return of −12.98 percent (*t*-statistic = −2.12). This finding supports the argument that investors pay a substantial premium to hedge against gamma risk only in the presence of concave IV curves.

Table 7 presents estimates from predictive panel regressions of JUMPSTRADDLE on CONCAVE and other controls. These regressions also control for the expiry and the average moneyness of each pair of options used to construct this calendar strategy, ensuring that the reported differential return is not driven by these features. Column (1) confirms that CONCAVE possesses significant predictive ability over JUMPSTRADDLE returns. Specifically, concave IV curves are followed by a 12.71 percent lower average JUMPSTRADDLE return on EADs, as compared to non-concave IV curves. This significant differential return is not subsumed when we control for additional firm characteristics and option-based risk measures in columns (2)–(5). In sum, concave IV curves signal the substantial premium investors are willing to pay to hedge against the gamma risk that arises due to earnings announcements.

5.3.3 Delta- and gamma-neutral straddle returns

To reinforce the argument that the informational content of concave IV curves is related to gamma rather than vega risk, we also construct delta- and gamma-neutral ATM straddles (VOLSTRADDLE) in a similar fashion. The first leg of this strategy is a long position in a delta-neutral ATM straddle constructed from options with the longest available expiry between 100 and 180 days. The second leg is a short position in Γ_L/Γ_S delta-neutral ATM straddles constructed from options with the shortest available expiry between 4 and

Table 7. Concave IV curves and delta- and vega-neutral straddle returns on EAD.
This table presents results from predictive panel regressions of delta- and vega-neutral ATM straddle returns (JUMPSTRADDLE) computed on EAD on CONCAVE and other variables. Stock controls, measured on the day prior to the EAD, in column (2) include BETA, Ln(SIZE), B/M, RUNUP, MOM, Ln(PRICE), DISP, and ANNBETA. The definition of all the variables is provided in Appendix A. All specifications include an intercept and two option controls, viz. the expiry and the average moneyness of the options used to construct this calendar straddle strategy. Coefficient estimates on ATMIV, RVIV, and TSIV are multiplied by 100 for easy readability. *t*-statistics in parentheses are calculated using two-way clustered standard errors at the firm- and quarter-level. The sample consists of quarterly earnings announcements during the period 2013–2020.

	(1)	(2)	(3)	(4)	(5)
CONCAVE	−12.71	−18.92	−14.04	−18.44	−17.44
	(−2.19)	(−3.10)	(−2.54)	(−3.03)	(−2.82)
ATMIV	—	—	−1.54	−92.35	−0.45
			(−0.10)	(−1.74)	(−0.03)
RVIV	—	—	—	−30.47	—
				(−1.26)	
TSIV	—	—	—	512.65	—
				(1.56)	
RNS	—	—	—	—	−18.02
					(−2.03)
RNK	—	—	—	—	−15.55
					(−3.76)
Stock Controls	No	Yes	No	No	No
# observations	1,882	1,660	1,862	1,852	1,862
R^2	0.42	1.21	0.48	0.88	1.04

13 days, where Γ_S (Γ_L) denotes the gamma of the shorter- (longer-) maturity straddle. This position ensures that this calendar strategy is gamma-neutral at formation. The VOLSTRADDLE return on EAD is given by:

$$\text{VOLSTRADDLE} = (1-w)R_S + wR_L, \tag{8}$$

where R_S (R_L) is the return of the shorter- (longer-) maturity delta-neutral straddle on EAD and $w = -(\Gamma_S/V_S)/(\Gamma_L/V_L - \Gamma_S/V_S)$, with V_S (V_L) denoting the value of the shorter- (longer-) maturity straddle at formation.

The average (median) VOLSTRADDLE return reported in Table 1 is 1.17 percent (0.81 percent). Distinguishing between concave and non-concave IV curves, Table 3 shows that the average VOLSTRADDLE return remains positive in both cases and the differential is insignificant. Hence, we conclude that investors do not pay a premium to hedge against the vega risk that may arise due to earnings announcements.

Table 8 confirms this finding in a predictive panel regression setup, controlling also for the expiry and the average moneyness of each pair of options used to construct this calendar strategy. Column (1) shows that concave IV curves are actually followed by a marginally higher, not lower, average VOLSTRADDLE return relative to non-concave IV curves, but this differential is insignificant. We get similar results when we control for other firm and option-based variables in columns (2)–(5). Overall, we conclude that vega risk is not significantly priced on EADs and that the observed concavity in IV curves is not related to this dimension of risk.

5.4 Portfolio analysis

The above regression analysis documents the significant predictive ability of CONCAVE with respect to STRADDLE, STRANGLE, and JUMPSTRADDLE returns controlling for other

Table 8. Concave IV curves and delta- and gamma-neutral straddle returns on EAD.

This table presents results from predictive panel regressions of delta- and gamma-neutral ATM straddle returns (VOLSTRADDLE) computed on EAD on CONCAVE and other variables. Stock controls, measured on the day prior to the EAD, in column (2) include BETA, Ln(SIZE), B/M, RUNUP, MOM, Ln(PRICE), DISP, and ANNBETA. The definition of all the variables is provided in Appendix A. All specifications include an intercept and two option controls, viz. the expiry and the average moneyness of the options used to construct this calendar straddle strategy. Coefficient estimates on ATMIV, RVIV, and TSIV are multiplied by 100 for easy readability. *t*-statistics in parentheses are calculated using two-way clustered standard errors at the firm- and quarter-level. The sample consists of quarterly earnings announcements during the period 2013–2020.

	(1)	(2)	(3)	(4)	(5)
CONCAVE	0.22	0.25	0.19	−0.03	0.02
	(1.44)	(1.64)	(1.08)	(−0.17)	(0.08)
ATMIV	—	—	0.17	−5.96	0.17
			(0.27)	(−5.63)	(0.28)
RVIV	—	—	—	0.58	—
				(0.46)	
TSIV	—	—	—	41.31	—
				(4.94)	
RNS	—	—	—	—	−0.07
					(−0.17)
RNK	—	—	—	—	−0.46
					(−3.54)
Stock Controls	No	Yes	No	No	No
# observations	1,889	1,665	1,869	1,858	1,869
R^2	0.45	1.37	0.47	2.08	0.91

option-based risk measures. Here, we further examine the economic significance of this finding using a portfolio analysis.

First, we construct univariate portfolio sorts on the basis of ATMIV, TSIV, and RVIV values and compute the corresponding STRADDLE, STRANGLE, and JUMPSTRADDLE premia on EADs. Specifically, we classify observations into a High or Low portfolio based on the respective median value in each quarter.

For each option strategy, the first column of Panel A in Table 9 reports the corresponding average portfolio returns as well as their spread using ATMIV as the sorting variable. We find that there are no significant differential STRADDLE, STRANGLE, and JUMPSTRADDLE premia between High and Low ATMIV portfolios. This is in stark contrast with the evidence reported in Table 3 (see also Tables 5, 6, and 7), according to which concave IV curves are associated with significantly lower STRADDLE, STRANGLE, and JUMPSTRADDLE premia relative to non-concave IV curves. Similarly, Panel B (Panel C) in Table 9 reports the corresponding results using TSIV (RVIV) as the sorting variable. We find that neither TSIV nor RVIV yields a significant spread for any of the three option strategies considered.[22]

In sum, this evidence confirms that ATMIV, TSIV, and RVIV are not significantly related to the premia of these option strategies on EADs. On the other hand, CONCAVE has a first-order effect on the premia of straddles and strangles earned on EADs, and hence on the pricing of gamma risk that arises due to earnings announcements.

We also construct bivariate portfolio sorts, where we firstly classify ATMIV, TSIV, or RVIV observations into a High or Low portfolio relative to their median quarter value, and then within each portfolio, we further classify observations based on whether they exhibit

[22] We have repeated this analysis alternatively constructing tercile portfolios. We similarly find that the differential STRADDLE, STRANGLE, and JUMPSTRADDLE average returns between High and Low ATMIV, TSIV, or RVIV portfolios are insignificant.

Table 9. Portfolio analysis.

This table presents the average returns of various option strategies computed on EADs for univariate and bivariate portfolio sorts. Each panel reports the average portfolio returns for delta-neutral ATM straddles (STRADDLE), delta-neutral strangles (STRANGLE), and delta- and vega-neutral ATM straddles (JUMPSTRADDLE). The first sorting variable is ATMIV in Panel A, TSIV in Panel B, and RVIV in Panel C. ATMIV, RVIV, and TSIV are defined in Appendix A. The classification of observations into a High or Low portfolio is based on the corresponding median quarter value. The first column in each subpanel shows average portfolio returns for the univariate sort. The second and third columns show the corresponding average returns for observations within the High or Low portfolio that exhibit concave (CONCAVE = 1) and non-concave IV curves (CONCAVE = 0) on the day prior to the EAD, respectively. The final column shows the difference between the second and third columns. *t*-statistics are provided in parenthesis. The sample consists of quarterly earnings announcements during the period 2013–2020.

	All	CONCAVE = 1	CONCAVE = 0	Difference
Panel A: Sorts on ATMIV				
		Delta-neutral straddles, STRADDLE		
High	−1.18	−2.65	0.49	−3.14 (−1.02)
Low	−0.67	−7.86	1.51	−9.37 (−3.00)
Difference	−0.51 (−0.24)			
		Delta-neutral strangles, STRANGLE		
High	−3.63	−6.35	−0.45	−5.90 (−1.11)
Low	−1.03	−13.38	2.53	−15.91 (−2.82)
Difference	−2.60 (−0.70)			
		Delta- and vega-neutral straddles, JUMPSTRADDLE		
High	−2.85	−8.28	3.47	−11.75 (−1.46)
Low	−3.11	−19.15	2.38	−21.54 (−2.23)
Difference	0.26 (0.04)			
Panel B: Sorts on TSIV				
		Delta-neutral straddles, STRADDLE		
High	−0.34	−1.44	1.01	−2.45 (−0.78)
Low	−1.30	−9.57	0.95	−10.52 (−3.32)
Difference	0.96 (0.46)			
		Delta-neutral strangles, STRANGLE		
High	−2.52	−5.67	1.42	−7.10 (−1.32)
Low	−1.87	−13.50	1.20	−14.71 (−2.51)
Difference	−0.65 (−0.18)			
		Delta- and vega-neutral straddles, JUMPSTRADDLE		
High	−2.86	−6.99	2.17	−9.16 (−1.21)
Low	−2.30	−18.39	2.84	−21.23 (−2.00)
Difference	−0.56 (−0.09)			
Panel C: Sorts on RVIV				
		Delta-neutral straddles, STRADDLE		
High	−2.12	−9.56	0.51	−10.07 (−3.44)
Low	0.26	−1.51	2.03	−3.54 (−1.14)
Difference	−2.38 (−1.13)			
		Delta-neutral strangles, STRANGLE		
High	−4.87	−19.60	0.37	−19.97 (−1.11)
Low	0.19	−2.62	2.97	−5.59 (−2.82)
Difference	−5.06 (−1.37)			
		Delta- and vega-neutral straddles, JUMPSTRADDLE		
High	−8.03	−26.46	−0.87	−25.59 (−2.71)
Low	1.99	−3.90	8.07	−11.97 (−1.51)
Difference	−10.01 (−1.58)			

concave (CONCAVE = 1) or non-concave IV curves (CONCAVE = 0) on the day prior to the EAD.

The second and third columns in each subpanel of Table 9 report the corresponding average portfolio returns for these bivariate sorts, whereas the last column contains the spread within each group. It is worth noting that all of these portfolios are well populated because CONCAVE is not highly correlated with either of the three option-based variables we examine (see Table 2).

We find that when CONCAVE = 0, STRADDLE, STRANGLE, and JUMPSTRADDLE premia are predominantly positive across ATMIV, TSIV, and RVIV portfolios. Hence, high ATMIV, high TSIV, or low RVIV values are *not* a sufficient condition for negative straddle or strangle premia on EADs. To the contrary, when CONCAVE = 1, these option strategies always yield negative premia across high and low ATMIV, TSIV, and RVIV portfolios.

We also find that within each ATMIV, TSIV, and RVIV group, the premium for CONCAVE = 1 observations is always lower (and negative) than the premium for CONCAVE = 0 observations. Interestingly, this spread appears to be economically more significant among low ATMIV, low TSIV, and high RVIV values. This evidence further supports the argument that the negative straddle and strangle premia we document in the presence of concave IV curves are not simply driven by high ATMIV, high TSIV, or low RVIV values prior to EADs; quite the opposite is true.

6. Beyond earnings announcements

6.1 Scheduled macroeconomic announcements

Our model and empirical methodology could be applied to any type of scheduled announcement or event as long as this is expected to trigger a substantial shift in the underlying asset price. For example, scheduled macroeconomic announcements and scheduled political events, such as elections and referendums, can give rise to phenomena similar to the ones we document around earnings announcements. In fact, there is a growing literature that examines the impact of macroeconomic announcements on stock premia (see, e.g., Savor and Wilson 2013; and a recent review of the literature by Ai, Bansal, and Guo 2023). Do IV curves of short-expiry options exhibit concavity (as an ex-ante signal of jumps) prior to these announcements?

To answer this question, we examine how prevalent the formation of concave IV curves is on the trading day prior to scheduled macroeconomic announcements during our sample period. Here, we exclude IV curves from options whose expiry spans EADs. In line with the prior literature, we consider the following three commonly examined categories: (1) FOMC announcements, (2) producer price index (PPI) announcements, and (3) Nonfarm Payroll Employment announcements. We obtain the announcement dates from the Federal Reserve and the Bureau of Labor Statistics, respectively. We report summary statistics for the fraction of concave IV curves observed prior to each announcement category in Table 10.

We find that, on average, the fraction of concave IV curves observed prior to macroeconomic announcements was similar to the one observed on a typical trading day. This is in line with our expectations because prior to 2020, our sample period covers a relatively calm macroeconomic regime (low inflation, declining unemployment rate, and stable GDP growth rate). Hence, prior to 2020, most of these announcements were not anticipated to have a substantial impact on stock prices. In fact, we find that the average excess market (close-to-close) return on announcement days across these categories was just 7 bps during the period 2013–2019. This is consistent with the evidence in Hu et al. (2022, Table 2).

Nevertheless, there is a substantial variation in the fraction of concave IV curves before macro announcements. In fact, we identify several announcements that were indeed preceded by a large fraction of concave IV curves. These announcements predominantly

Table 10. Fraction of concave IV curves prior to scheduled macroeconomic announcements.
This table presents summary statistics for the fraction (in percent) of concave IV curves on selected days. IV curves are computed from short-expiry options with time to maturity between three and thirteen calendar days, excluding options whose expiry spans earnings announcements. These fractions are computed on the trading day prior to FOMC announcements, Nonfarm Payroll Employment announcements, and PPI announcements, respectively. The last row shows the corresponding summary statistics for all trading days in our sample period, 2013–2020.

	Mean	St. Dev.	25th pctl	Median	75th pctl	No. of days
FOMC announcements	3.61	5.13	0	1.75	5.08	63
Employment announcements	3.89	4.74	0	3.06	5.05	96
PPI announcements	4.22	5.75	0	2.20	5.49	96
All trading days	4.56	6.57	0	2.78	6.25	2,015

occurred during 2020, but there are cases such as these in the prior period as well.[23] Collectively, our results show that the anticipation of stock price jumps, in either direction, is an important component of the heightened uncertainty prior to scheduled announcements, which is shown to be priced (Hu et al. 2022; Gao, Hu, and Zhang 2023).

6.2 Concave IV curves on non-EAD

We further examine the time-series behavior of the aggregate fraction of concave IV curves excluding those that span EADs. We start by noting that this fraction exhibits substantial time-series variation. The median daily fraction of concave IV curves is just 2.78 percent, whereas the corresponding average daily fraction is 4.56 percent. This feature reveals that the distribution of these fractions is positively skewed, with certain days in our sample period exhibiting a very large fraction of concave IV curves. At the same time, there are many trading days in which the fraction of concave IV curves is 0 percent.

Figure 10 illustrates the fraction of concave IV curves (blue line, left axis) aggregated at the monthly frequency. We observe that the fraction of concave IV curves was particularly high from March 2020 onward. Prior to 2020, we also observe spikes on certain other months (e.g., August 2015, January 2016, February 2018, and December 2018). To understand this time-series behavior, recall that concavity in the IV curve is an ex-ante signal for anticipated jumps in the underlying stock price under the risk-neutral measure $\mathbb{Q}$. Therefore, concavity tends to be observed when stock price jumps are more likely to occur under the physical measure $\mathbb{P}$ and/or when investors overstate the probability of jumps as we move from $\mathbb{P}$ to $\mathbb{Q}$ due to their preferences or beliefs. For the case of earnings announcements, we have shown that this feature is primarily driven by the very high incidence of stock price jumps on EADs under the $\mathbb{P}$ measure. Do concave IV curves excluding EADs also reflect ex-ante anticipation of stock price jumps?

To answer this question, figure 10 also illustrates the incidence of stock price jumps for the same sample of firms. Specifically, for the firm-day observations we extract an IV curve from options whose expiry does not span an EAD, we compute the fraction of stock price jumps observed on the next trading day. The indicator variable JUMP2 (JUMP3) takes the value 1 when the absolute stock return on day t is at least two (three) times higher than its past realized volatility computed from $t-250$ to $t-25$. Figure 10 aggregates the fraction of realized stock price jumps at the monthly frequency.

It is evident from figure 10 that the fraction of concave IV curves aligns strongly with the incidence of stock price jumps. The most obvious coincidence of these two series occurs at

[23] As an illustrative example, prior to FOMC announcements, the largest fraction of concave IV curves (24.6 percent) was observed on Nov 4, 2020, that is, the day after the US presidential election (Nov 3) and prior to the Fed's announcement on Nov 5. The Fed kept interest rates near zero and Chairman Powell expressed the Fed's strong commitment to use monetary policy to support the economy; on that announcement day the S&P 500 rose by 1.95 percent.

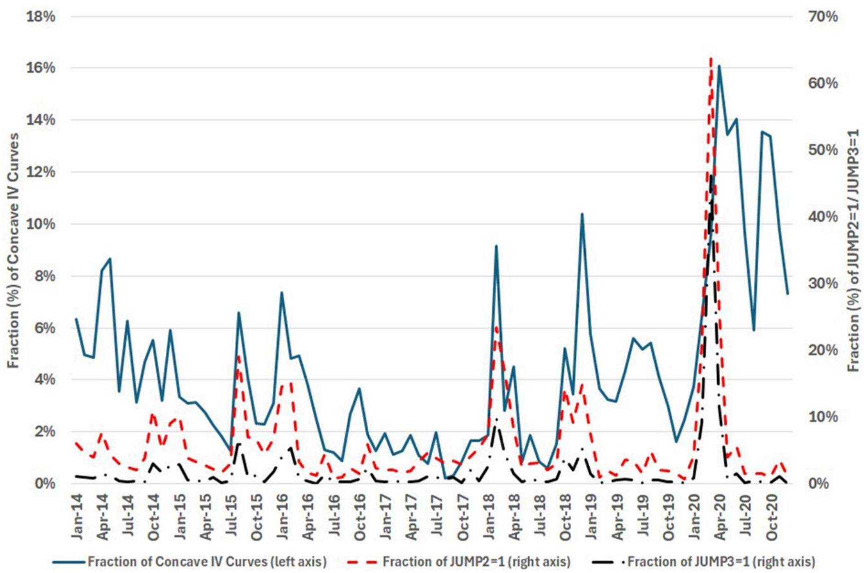

Figure 10. Fractions of concave IV curves and realized stock price jumps excluding earnings announcements. This figure shows the fraction of concave IV curves (blue line, left axis) aggregated at the monthly frequency. IV curves are extracted from short-maturity equity options, whose expiry does not span the firm's quarterly earnings announcement. For this set of extracted IV curves, the figure also shows the fraction of stock price jumps realized on the following trading day. JUMP2 (JUMP3) is an indicator variable that takes the value 1 when the absolute stock return on day t is at least two (three) times higher than its past realized volatility computed from $t-250$ to $t-25$. The fraction of JUMP2 = 1 (JUMP3 = 1) observations aggregated at the monthly frequency is shown by the red dash (black dash-dot) line corresponding to the right axis. The sample period for this figure is January 2014 to December 2020.

the onset of the pandemic crisis and the corresponding economic meltdown in Spring 2020. Before this period, spikes in the incidence of stock price jumps also coincide with spikes in the fraction of concave IV curves observed in August 2015, January 2016, February 2018, and December 2018. Overall, the fraction of concave IV curves is significantly correlated with the fraction of JUMP2 (JUMP3), exhibiting a correlation coefficient of 0.335 (0.291) during the entire sample period.

Thus, even when we exclude earnings announcements, concave IV curves primarily reflect the anticipation of subsequently realized stock price jumps. We do note that there are also months in our sample when high fractions of concavity were not followed by a similarly high incidence of jumps. An example of this situation is the particularly turbulent period of September to October 2020, which was characterized by the combination of the ongoing pandemic crisis and the heightened political uncertainty prior to the US presidential election. This example shows that a high fraction of concave IV curves can also appear during turbulent periods when investors are willing to pay a very high premium to hedge against gamma risk.

6.2.1 Aggregate uncertainty or differences in opinion?

To further examine the economic implications of this time-series behavior, we analyze whether the incidence of concavity excluding earnings announcements more likely reflects aggregate uncertainty or differences of opinion. We find that the ratio of concave IV curves is strongly positively correlated with the macroeconomic and financial uncertainty indices of Jurado, Ludvigson and Ng (2015) and Ludvigson, Ma, and Ng (2021), with correlation coefficients equal to 0.616 and 0.556, respectively. The strong positive correlation with the macroeconomic uncertainty index is particularly interesting, as the latter is constructed from a broad set of macroeconomic times series, mostly measuring real economic activity, and it does not simply reflect market volatility. This strong correlation also implies that there is an increased anticipation of stock price jumps during uncertain macroeconomic times.

In contrast, the fraction of concave IV curves is only weakly correlated with the analyst forecast disagreement measure of Yu (2011), which derives from the value-weighted dispersion of analyst forecasts with respect to firms' earnings per share long-term growth rate. The corresponding correlation coefficient is only 0.097. This low correlation at the aggregate level is consistent with the negligible correlation between CONCAVE and DISP that we report in Table 2 prior to earnings announcements, and the absence of bimodality in analyst forecasts (footnote 10). Together, this correlation analysis shows that the concavity in the IV curve mainly reflects a higher degree of uncertainty rather than differences in opinion.[24]

Finally, we find that the fraction of concave IV curves is strongly negatively correlated with the Intermediary Capital Ratio (ICR) of He, Kelly, and Manela (2017), with a coefficient of –0.631. Consistent with this negative correlation, adverse financial market conditions (e.g., as in March to April 2020) are typically associated with both a lower capital ratio for intermediaries and a higher incidence of stock price jumps, the anticipation of which is reflected into a higher fraction of concave IV curves. Another reason for this negative correlation is that financial intermediaries become constrained when their capital ratio substantially drops, and hence require a substantial premium to sell options with high exposure to gamma risk, such as short-expiry straddles. This is manifested by substantially higher implied volatilities for these options, which results in the formation of concave IV curves. This mechanism may explain the coincidence of a very low ICR and a high fraction of concave IV curves in September to October 2020, despite the relatively low incidence of stock price jumps during that period.

7. Conclusion

We document that the IV curves of equity options frequently exhibit concavity prior to the EAD. This shape is in stark contrast with the convex volatility smiles or smirks that are commonly observed for equity options. Concavity is most obvious in short-expiry options, it typically reflects a bimodal RND for the underlying stock price, and quickly disappears after the announcement, as the uncertainty surrounding this event is resolved.

We report evidence that firms with concave IV curves exhibit higher absolute abnormal stock returns on EAD and higher realized volatility after the announcement. Despite the larger than average stock price moves on EAD following the formation of concave IV curves, we still find that the corresponding delta-neutral straddle returns are significantly lower than those for non-concave IV curves. To rationalize this finding, we show that ATM straddles are significantly more expensive in the presence of concave IV curves, and hence the realized stock price jumps are not sufficient to offset the substantial cost of these straddles. We further show that concave IV curves are followed by large negative strangle and delta- and vega-neutral straddle returns on EADs, revealing that investors seek to hedge the gamma, rather than vega risk that arises due to this corporate event.

Overall, we show that investors can ex-ante identify the announcements that trigger larger than average stock price moves and they pay a substantial premium to hedge against this event risk. This hedging activity impacts on option prices, leading to the formation of a

[24] We do find that the fraction of concave IV curves is strongly positively correlated with the monthly average of daily VIX values, exhibiting a correlation coefficient of 0.707. This strong correlation at the aggregate level confirms the mechanics of our model, where anticipation of stock price jumps in either direction both increases the level of implied volatility and generates concavity in the IV curve of short-expiry options.

concave IV curve. We conclude that concavity in the IV curve constitutes an ex-ante option-implied signal for event risk in the underlying stock arising due to the impending announcement.

The focus of our study is on scheduled corporate earnings announcements. However, our model as well as the empirical methodology can be applied to any type of scheduled announcement that may trigger large asset price moves. While prior studies have argued that macroeconomic announcements and geopolitical events can give rise to substantial risk, which can be ex-ante reflected in option prices, our results additionally show that the curvature properties of the IV curve can provide additional information with respect to the pricing of event risk and the timing of the resolution of uncertainty that arises due to these announcements.

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Supplementary material

Supplementary material is available at *Review of Finance* online.

Conflicts of interest: None declared.

Data availability

The data underlying this article were provided under license by CRSP, Compustat, and OptionMetrics. The datasets were purchased through our institutions. Scrambled data and code are provided at Harvard Datavserse.

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Appendix

Appendix A: Definition of variables

ABSEADABRET: We compute the abnormal stock return on EAD as the realized minus the expected return. The expected return is calculated on the basis of pre-estimated factor loadings for each firm using the FFC four-factor model. For this estimation, we use daily returns from $d-250$ to $d-25$, where d is the EAD, requiring at least 200 observations. This choice ensures that the estimated factor loadings are not affected by stock returns observed in the runup to the EAD.

ANNBETA: Following Barth and So (2014), announcement beta is the estimate of coefficient β_3 from the following firm-level regression model:

$$xr_{i,t} = \alpha_i + \beta_{1,i}MKT_t + \beta_{2,i}AnnDay_{i,t} + \beta_{3,i}(MKT_t \times AnnDay_{i,t}) + \varepsilon_{i,t}, \tag{A.1}$$

where $xr_{i,t}$ is the excess daily return of stock i on day t, MKT denotes the excess market return, and $AnnDay_{i,t}$ is an indicator variable that takes the value 1 on trading days $(d-1, d, d+1)$, where d is the EAD, and 0 otherwise. We estimate this model using daily data during the past twelve quarters. We require at least 8 EADs and at least 451 observations.

`ATMIV`: The average of the annualized call IV with $\Delta_{CALL} = 0.5$ and the annualized put IV with $\Delta_{PUT} = -0.5$. Annualized implied volatilities are from the 10-day Volatility Surface File of OptionMetrics.

`B/M`: The ratio of firm book value of equity (CEQ) to market capitalization. Market capitalization is defined as the product of share price (PRC) times the number of shares outstanding (SHROUT). We drop observations with negative book value. We use the `B/M` computed at the end of the previous fiscal quarter.

`BETA`: The market beta estimated from the FFC four-factor model. We estimate this model at t using daily data from $t-250$ to $t-25$ and requiring at least 200 observations. *MKT*, *SMB*, *HML*, and *WML* returns are from Kenneth French's online data library.

`DISP`: The standard deviation of the earnings per share (EPS) forecasts for the next quarterly earnings announcement scaled by the absolute value of the mean EPS forecast. EPS forecasts are from I/B/E/S.

`Ln(PRICE)`: The natural logarithm of the share price (PRC).

`Ln(SIZE)`: The natural logarithm of the firm's market capitalization (in million \$). Market capitalization is defined as the product of share price (PRC) times the number of shares outstanding (SHROUT). We use the market capitalization computed at the end of the previous fiscal quarter.

`MOM`: The cumulative stock return from day $t-250$ to day $t-25$. We require at least 200 daily observations.

`NUMEST`: The number of analysts providing EPS forecasts for the next quarterly earnings announcement is from I/B/E/S.

`O/S`: The ratio of daily option trading volume to daily stock trading volume. Option trading volume is multiplied by 100, as each option contract corresponds to a 100-share lot. We sum up the trading volume of all call and put options with the same expiry as the one used to define the indicator `CONCAVE`.

`POSTEADVOL`: We compute the annualized 10-day stock return volatility from d to $d+9$, where d is the EAD, according to the standard formula:

$$\texttt{POSTEADVOL} = \sqrt{\frac{252}{10}\sum_{t=d}^{d+9} r_t^2} \tag{A.2}$$

where r_t is the daily log-return.

`RVIV`: The difference between the annualized realized (historical) volatility and `ATMIV`. Realized volatility is from the 10-day Historical Volatility File provided by OptionMetrics.

`RUNUP`: The cumulative stock return from day $t-4$ to day t. We require all five daily observations.

RNK: The risk-neutral kurtosis computed as per the definition of Bakshi, Kapadia, and Madan (2003). We use prices of OTM and ATM options with the same expiry as the one used to define the indicator CONCAVE. We require at least four options, with at least two calls and two puts. Option prices are converted to implied volatilities and vice versa via the Black–Scholes formula. We use a cubic spline to interpolate implied volatilities between the lowest and the highest available strikes and perform a constant extrapolation outside this range, with lower bound $K/S = 1/3$ and upper bound $K/S = 3$.

RNS: The risk-neutral skewness computed as per the definition of Bakshi, Kapadia, and Madan (2003). We use prices of OTM and ATM options with the same expiry as the one used to define the indicator CONCAVE. We require at least four options, with at least two calls and two puts. Option prices are converted to implied volatilities and vice versa via the Black–Scholes formula. We use a cubic spline to interpolate implied volatilities between the lowest and the highest available strikes and perform a constant extrapolation outside this range, with lower bound $K/S = 1/3$ and upper bound $K/S = 3$.

STOCKTR: The ratio of daily stock trading volume to shares outstanding.

TSIV: The term structure estimator of ATM IV proposed by DJKS (2019) and defined as the square root of the following expression:

$$(\sigma^{\mathbb{Q}}_{i,term})^2 = \frac{\sigma^2_{t,T_1} - \sigma^2_{t,T_2}}{T_1^{-1} - T_2^{-1}}, \tag{A.3}$$

where σ^2_{t,T_1} is the squared annualized ATM IV corresponding to the nearest expiry T_1 and σ^2_{t,T_2} is the squared annualized ATM IV corresponding to the second nearest expiry T_2. T_1 is the same as the maturity of the options used to define the indicator CONCAVE. We use the nearest-to-the-money option to compute the ATM IV, with moneyness defined as the strike price divided by the forward price. TSIV is not defined when $\sigma^2_{t,T_1} < \sigma^2_{t,T_2}$.

Appendix B: Characteristic function

Let $\varphi(u; t, T, S_t, V_t)$ with $u \in \mathbb{R}$ be the characteristic function of $\log S_T$ conditional on $\mathcal{F}_t$ under risk-neutral measure $\mathbb{Q}$. According to our model, the log stock price is the sum of two independent components. The first component is an affine process for which the characteristic function, denoted as $\varphi_{af}(u; t, T, S_t, V_t)$, is known in closed-form (see Bates 1996) and it is given by:

$$\varphi_{af}(u; t, T, S_t, V_t) = \exp\Big(iu\ln S_t + \alpha(u; t, T) + \beta(u; t, T)V_t\Big), \tag{B.1}$$

where

$$\begin{aligned} \alpha(u; t, T) &= (r - \lambda_J\bar{\mu}_J)\tau iu + \frac{\kappa_v\theta_v}{\sigma_v^2}\left(q_1\tau - 2\log\left(\frac{1 - ge^{\Delta\tau}}{1 - g}\right)\right) \\ &\quad + \lambda_J\tau\Big((1 + \bar{\mu}_J)^{iu} e^{\frac{\sigma_J^2}{2}iu(iu-1)} - 1\Big) \\ \beta(u; t, T) &= \frac{q_1}{\sigma_v^2}\left(\frac{1 - e^{\Delta\tau}}{1 - ge^{\Delta\tau}}\right), \end{aligned} \tag{B.2}$$

with $\tau = T - t$, $\Delta = \sqrt{(\kappa_v - \rho\sigma_v iu)^2 - 2\sigma_v^2 iu(iu - 1)}$, $q_1 = \kappa_v - \rho\sigma_v iu + \Delta$, $q_2 = \kappa_v - \rho\sigma_v iu - \Delta$ and $g = q_1/q_2$.

The second component is a discrete process with independent deterministic jumps at known times. Its characteristic function, also known in closed-form, is given by:

$$\varphi_{dis}(u;t,T) = \prod_{j=N_t^d+1}^{N_T^d} \varphi_j(u), \tag{B.3}$$

where

$$\varphi_j(u) = p_j M_j^{(-)}(u) + (1-p_j) M_j^{(+)}(u), \tag{B.4}$$

where $M_j^{(-)}(u) = \exp\left(iu\mu_j^{(-)} + \frac{(iu)^2}{2}\left(\sigma_j^{(-)}\right)^2\right)$ and $M_j^{(+)}(u) = \exp\left(iu\mu_j^{(+)} + \frac{(iu)^2}{2}\left(\sigma_j^{(+)}\right)^2\right)$.

As the two components are independent, $\varphi(u;t,T,S_t,V_t)$ is given by the product of the characteristic functions (B1) and (B3):

$$\varphi(u;t,T,S_t,V_t) = \varphi_{af}(u;t,T,S_t,V_t)\varphi_{dis}(u;t,T). \tag{B.5}$$

Knowledge of the characteristic function of the log stock price in closed-form enables us to price options. In particular, the price at time t of a European call option with strike price K and expiry T, denoted as $C_t(K,T)$, is given by (see Heston and Nandi 2000):

$$\begin{aligned} C_t(K,T) = {} & \frac{1}{2}S_t + \frac{e^{-r(t-t)}}{\pi}\int_0^\infty \mathrm{Re}[\frac{K^{-iu}\varphi(u-i;t,T,S_t,V_t)}{iu}]du \\ & - Ke^{-r(T-t)}\left(\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}[\frac{K^{-iu}\varphi(u;t,T,S_t,V_t)}{iu}]du\right). \end{aligned} \tag{B.6}$$

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